ETSI TS 124 571 V19.5.0 (2026-09)



TECHNICAL SPECIFICATION

5G;
5G System (5GS);
Control plane Location Services (LCS) procedures;
Stage 3
(3GPP TS 24.571 version 19.5.0 Release 19)

Reference
RTS/TSGC-0124571vj50

Keywords
5G

ETSI

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Foreword

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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the operations and information coding for the Non-Access Stratum (NAS) layer protocol for supporting the Location Services (LCS) in the NG-RAN.

The present document is applicable to the User Equipment (UE) and to the Access and Mobility Management Function (AMF) or Location Management Function (LMF) in the 5G System (5GS).

The present document is developed in accordance to the general principles stated in 3GPP TS 23.273 [2].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".

[3] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".

[4] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".

[4a] 3GPP TS 38.355: "Sidelink Positioning Protocol (SLPP); Protocol specification".

[5] 3GPP TS 24.080: "Mobile radio interface layer 3 supplementary services specification; Formats and coding".

[6] 3GPP TS 29.572: "5G System; Location Management Services; Stage 3".

[7] 3GPP TS 38.305: "Stage 2 functional specification of User Equipment (UE) positioning in NG-RAN".

[8] 3GPP TS 22.261: "Service requirements for the 5G system; Stage 1".

[9] 3GPP TS 24.572: "User Plane Location Services (LCS) Protocols And Procedures; Stage 3".

[10] 3GPP TS 23.586: "Architectural Enhancements to support; Ranging based services and Sidelink Positioning".

[11] 3GPP TS 38.455: "NG-RAN; NR Positioning Protocol A (NRPPa)".

[12] 3GPP TS 23.003: "Numbering, addressing and identification".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

ExtendedFacility IE for 5G LCS: Usage of ExtendedFacility information element in the LCS operations as specified in clause 5.

For the purposes of the present document, the following terms and definitions given in 3GPP TS 22.261 [8] apply:

satellite NG-RAN

PRU ON/OFF state: This state is defined for a UE associated as a PRU with an LMF (i.e. after the successful completion of PRU association procedure). This state indicates whether the PRU functionality of the UE is activated (i.e. ON state) or deactivated (i.e. OFF state) while it is associated with an LMF.

UE operating as PRU: From the UE perspective, a UE which supports PRU functionality and operates as PRU. From the network perspective, the UE allowed to operate as a PRU based on the UE subscription data and the PRU state is ON.

3.2 Symbols

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5GCN	5G Core Network
5GS	5G System
AMF	Access and Mobility Management Function
DL	Downlink
IE	Information Element
GMLC	Gateway Mobile Location Centre
LCS	Location Services
LDR	Location Deferred Request
LMF	Location Management Function
LPP	LTE Positioning Protocol
MO-LR	Mobile Originated Location Request
MT-LR	Mobile Terminated Location Request
NAS	Non-access Stratum
PRU	Positioning Reference Unit
RSPP	Ranging and sidelink positioning protocol
SLPP	SideLink Positioning Protocol
SNPN	Stand-alone Non-Public Network
UE	User Equipment
UL	Uplink

4 General

4.1 Overview

4.1.1 NAS aspect

To enable transfer of Location Services (LCS) signaling messages between the 5G core network (5GCN) and the UE, two Payload container types are defined in the downlink (DL NAS TRANSPORT message) and the uplink (UL NAS TRANSPORT message or CONTROL PLANE SERVICE REQUEST message). The message protocol and procedures are described in 3GPP TS 24.501 [3].

4.1.2 LCS aspect

LCS uses the defined payload container to transfer LCS signalling messages between the UE and the network.

The corresponding LCS signaling messages include:

a) LTE Positioning Protocol (LPP) messages (see 3GPP TS 37.355 [4]):

- Both downlink and uplink LPP messages are supported.
- Routing information is transported as the Additional information IE in UL NAS TRANSPORT message or DL NAS TRANSPORT message for LPP messages (see 3GPP TS 24.501 [3]).

a1) SideLink Positioning Protocol (SLPP) messages (see 3GPP TS 38.355 [4a]):

- Both downlink and uplink SLPP messages are supported.
- Routing information is transported as the Additional information IE in UL NAS TRANSPORT message or DL NAS TRANSPORT message for SLPP messages (see 3GPP TS 24.501 [3]).

b) Location services messages:

- Messages for lcs-MOLR operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-LocationNotification operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-EventReport operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-PeriodicTriggeredInvoke operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-CancelDeferredLocation operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-MSCancelDeferredLocation operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-LocationPrivacySetting operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-PruAssociation operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-PruDisassociation operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-SLMOLR operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-SLMTLR operations (see 3GPP TS 24.080 [5]).
- Routing information associated with the LMF is transported as the Additional information IE in UL NAS TRANSPORT message or DL NAS TRANSPORT message or CONTROL PLANE SERVICE REQUEST message (see clause 5.2.2.6.1) for Location services messages that are transported between the UE and the LMF (see 3GPP TS 24.501 [3]).

c) RSPP transport messages:

- Messages for lcs-ULRSPPTransport operations (see 3GPP TS 24.080 [5]).
- Messages for lcs-DLRSPPTransport operations (see 3GPP TS 24.080 [5]).

The Routing information transported as the Additional information IE can be Routing identifier in the current specification, which includes immediate routing identifier and deferred routing identifier, which have LCS session level granularity (i.e. these identifiers are allocated per each LCS session). The immediate routing identifier transported as the Additional information IE is the LCS correlation identifier, which is allocated by the AMF and can be used in the UL NAS TRANSPORT message or DL NAS TRANSPORT message (see clause 5.2 and clause 5.3). The deferred routing identifier transported as the Additional information IE can be used in the UL NAS TRANSPORT and CONTROL PLANE SERVICE REQUEST message for the AMF routing the LCS messages to the particular LMF (see clause 5.2.2).

The messages for lcs-PeriodicTriggeredInvoke operations can be used to enable the location events reporting over user plane connection, and the messages for lcs-EventReport operations can be used for the cumulative event report, as

described in clause 6.16.1 of 3GPP TS 23.273 [2]. The location events reporting over user plane connection is defined in 3GPP TS 24.572 [9]).

4.2 LCS Support capabilities

4.2.1 UE support of LCS

The UE announces to the network its ability to support:

a) LCS notification mechanism;

b) LPP messages;

c) ExtendedFacility IE for 5G LCS; or

d) any combination of a, b and c;

using the 5GMM capability IE defined in 3GPP TS 24.501 [3].

The UE supporting LCS notification mechanism and LPP messages may support PRU (defined in 3GPP TS 38.305 [7]). The UE's ability to support PRU is stored in the UDM (see 3GPP TS 23.273 [2]) without any indication in 5GMM capability IE defined in 3GPP TS 24.501 [3].

A UE accessing 5G network via satellite NG-RAN shall not operate as a PRU.

4.2.2 Network support of LCS

The network announces to the UE its ability to support LCS in 5GC using the 5GS network feature support IE defined in 3GPP TS 24.501 [3]. The information is taken into account by the UE, in addition to UE's LCS capabilities, for the UE to determine whether to initiate MO-LR procedure in NG-RAN as specified in 3GPP TS 23.273 [3], clause 6.2.

If the ExtendedFacility IE for 5G LCS is supported homogeneously across all LMFs and is supported by the AMF, the network announces to the UE its ability to support ExtendedFacility IE for 5G LCS using the 5GS network feature support IE defined in 3GPP TS 24.501 [3].

NOTE: An SNPN can provide Location Services except for the cases of interworking with EPC, and roaming, which are not supported in an SNPN.

5 Support of LCS signalling

5.1 General

This clause defines the NG-RAN LCS operations (clause 5.2) and the format and coding of the messages (clause 5.3).

The messages defined in this clause can be transported in the UL NAS TRANSPORT message or DL NAS TRANSPORT message defined in 3GPP TS 24.501 [3].

5.2 LCS operations

5.2.1 Network initiated location services operations

5.2.1.1 Supplementary Services Location Notification

5.2.1.1.1 General

The supplementary services Location Notification operation enables the AMF to trigger the end-user notification verification process on the UE using NAS signalling. The supplementary services messages are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. UE Location Privacy Indication information may be included in Location Notification Return Result to indicate whether subsequent LCS requests or ranging and sidelink positioning requests will be allowed or disallowed by the UE. Figure 5.2.1.1.1.1 illustrates an example of the NAS signalling transport applicable to a 5GC-MT-LR procedure for immediate or deferred location or to an SL-MT-LR procedure for immediate or deferred location.

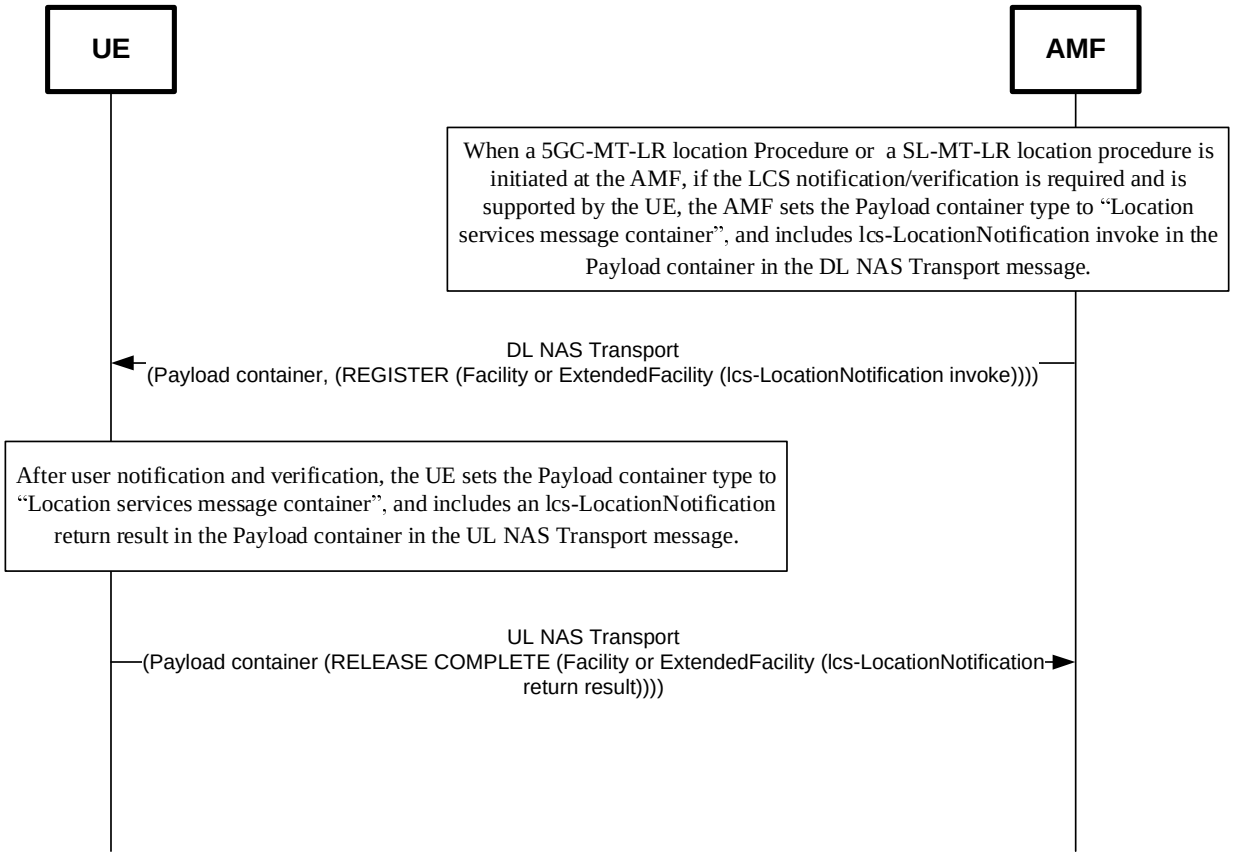


Figure 5.2.1.1.1.1: NAS signalling transport for Supplementary Services Location Notification

NOTE: The optional Additional information IE of the DL NAS TRANSPORT message or UL NAS TRANSPORT message is not included when the LCS Location Notification signalling is transported in the Payload container.

5.2.1.1.2 Normal operation

The network invokes a location notification procedure by sending a REGISTER message containing an lcs-LocationNotification invoke component to the UE as defined in 3GPP TS 24.080 [5]. This may be sent either to request verification for a 5GC-MT-LR or to notify the UE about an already authorized 5GC-MT-LR.

In case of privacy verification for LCS or ranging and sidelink positioning, the UE shall respond to the request by sending a RELEASE COMPLETE message containing the user's response and optionally UE Location Privacy Indication information in a lcs-LocationNotification return result component (see figure 5.2.1.1.2.1). If rangingSlExt is

included, the privacy verification is for ranging and sidelink positioning and the corresponding verification result for ranging and sidelink positioning is included in UE Location Privacy Indication information.

During the supplementary services Location Notification operation, the network shall run the timer T(LCSN) defined in 3GPP TS 24.080 [5]. This timer is started when the operation is sent, and stopped when a response is received from the UE. If the timer T(LCSN) expires in the network before any response from the UE (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in 3GPP TS 23.273 [2] (i.e. disallow location if barred by subscription or allow location if allowed by subscription). The network may also terminate the dialogue by sending a RELEASE COMPLETE message.

In the case of location notification, the UE shall terminate the dialogue immediately by sending a RELEASE COMPLETE message containing a lcs-LocationNotification return result.

If the UE is unable to process the request received from the network, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

The use of Facility or ExtendedFacility IE in figure 5.2.1.1.2.1 is specified in clause 5.3.1.

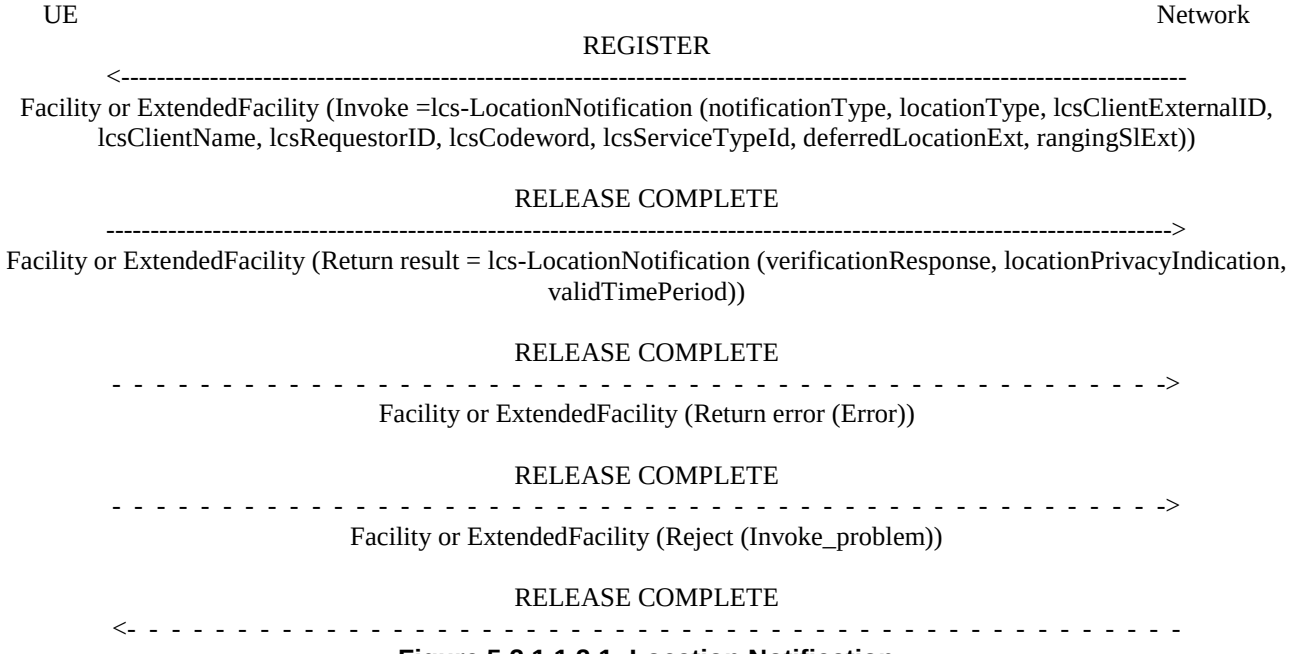


Figure 5.2.1.1.2.1: Location Notification

5.2.1.2 Positioning Information Transport

The AMF sends an LPP message and a Routing identifier which is the associated LCS correlation identifier in the DL NAS TRANSPORT message (refer to 3GPP TS 24.501 [3] and 3GPP TS 23.273 [2] clause 6.11.1) to a UE, or a UE operating as PRU, or both. Figure 5.2.1.2.1 illustrates an example of the NAS signalling transport for downlink LPP messages.

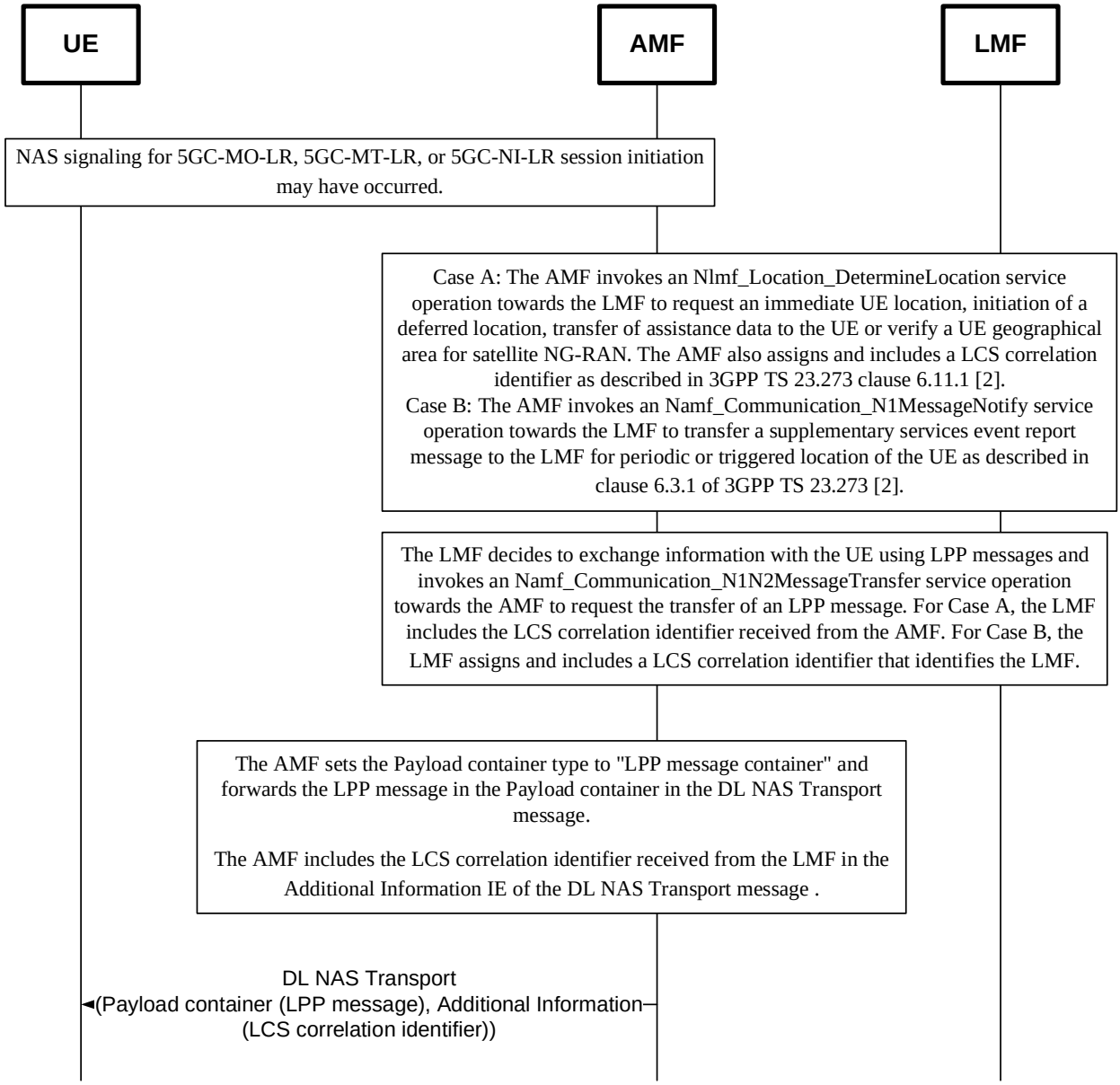


Figure 5.2.1.2.1: NAS signalling transport for downlink LPP messages

NOTE 1: If a scheduled location time is received in the LPP message, the UE in 5GMM-IDLE mode can initiate a service request procedure shortly before the scheduled location time.

NOTE 2: If the LMF determines that simultaneous measurements for UE and PRU(s) are needed, the LMF may send the time window(s) rather than the scheduled location time in the LPP messages to the UE, PRU(s) and NG-RAN. Definition of the time window and the associated configuration parameters are specified in TS 37.355 [4] and TS 38.455 [11].

5.2.1.3 Supplementary Services Periodic or Triggered Location

5.2.1.3.1 General

The supplementary services LCS PeriodicTriggered Invoke operation enables the LMF to initiate periodic or triggered location event reporting by a target UE as described in clause 6.3.1 and clause 6.19.2.1 of 3GPP TS 23.273 [2]. lcs-PeriodicTriggeredInvoke invoke component is transferred to the target UE via the serving AMF in a DL NAS TRANSPORT message. A response from the target UE is similarly returned to the LMF via the serving AMF and is transferred to the AMF in an UL NAS TRANSPORT message. If the lcs-PeriodicTriggeredInvoke invoke component in the Payload container IE of a DL NAS TRANSPORT message includes the deferred routing identifier, the UE shall

include the deferred routing identifier in the Additional information IE of an UL NAS TRANSPORT message during the subsequent procedures as specified in clause 5.2.2.

Figure 5.2.1.3.1.1 illustrates an example of the NAS signalling transport for initiation of periodic or triggered location.

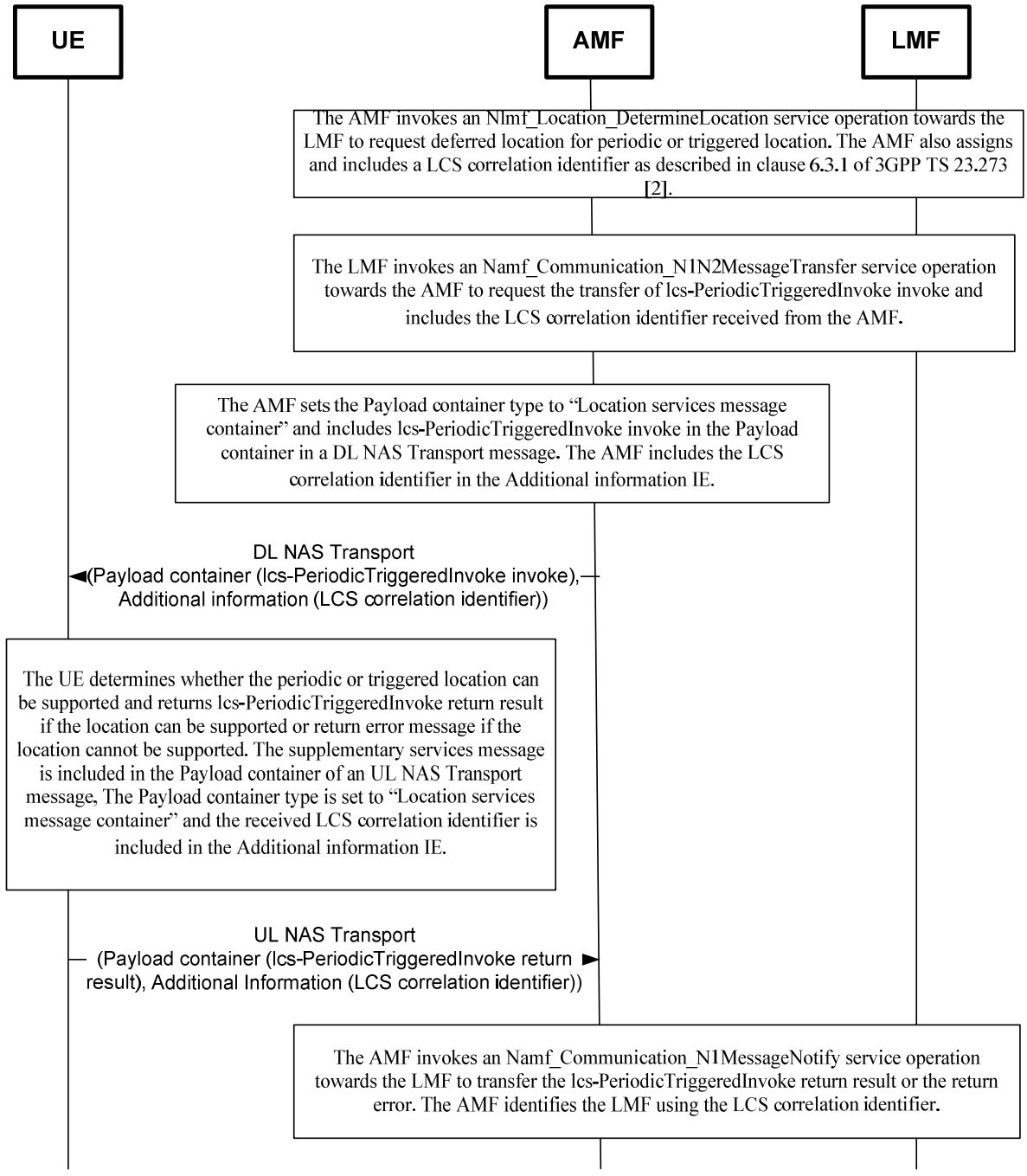


Figure 5.2.1.3.1.1: NAS signalling transport for LCS PeriodicTriggered messages

NOTE: If a scheduled location time is received in the lcs-PeriodicTriggeredInvoke invoke component, the UE in 5GMM-IDLE mode can initiate a service request procedure shortly before the scheduled location time.

5.2.1.3.2 Normal operation

The LMF sends a REGISTER message to the UE containing lcs-PeriodicTriggeredInvoke invoke component as defined in 3GPP TS 24.080 [5]. The REGISTER message is transported to the UE via the serving AMF as described in figure 5.2.1.3.1.1.

If the UE can support the periodic or triggered location, the UE returns a RELEASE COMPLETE message to the LMF containing an lcs-PeriodicTriggeredInvoke return result component. The RELEASE COMPLETE message is transported to the LMF via the serving AMF as described in figure 5.2.1.3.1.1.

For supporting of the location events reporting over user plane connection, as described in clause 6.16.1 of 3GPP TS 23.273 [2], the LMF provides the UE with the following:

a) the endpoint address for the location reporting over user plane connection;

b) the security information for the location reporting over user plane connection;

c) the cumulative event report timer, if available; or

d) the maximum number of events reporting counter, if available.

If the UE receives the cumulative event report timer or the maximum number of events reporting counter from the LMF, the UE shall perform the UE initiated Event Reporting procedure as defined in clause 5.2.2.4 with including the indication for the cumulative event report.

If the UE is unable to process or support the request received from the network, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component or reject component. Error values are specified in 3GPP TS 24.080 [5].

During the supplementary services periodic or triggered location operation, the network shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the UE. If this timer expires the network shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message.

Figure 5.2.1.3.2.1 illustrates the signalling for normal operation between the UE and the network. The use of Facility or ExtendedFacility IE is specified in clause 5.3.1.

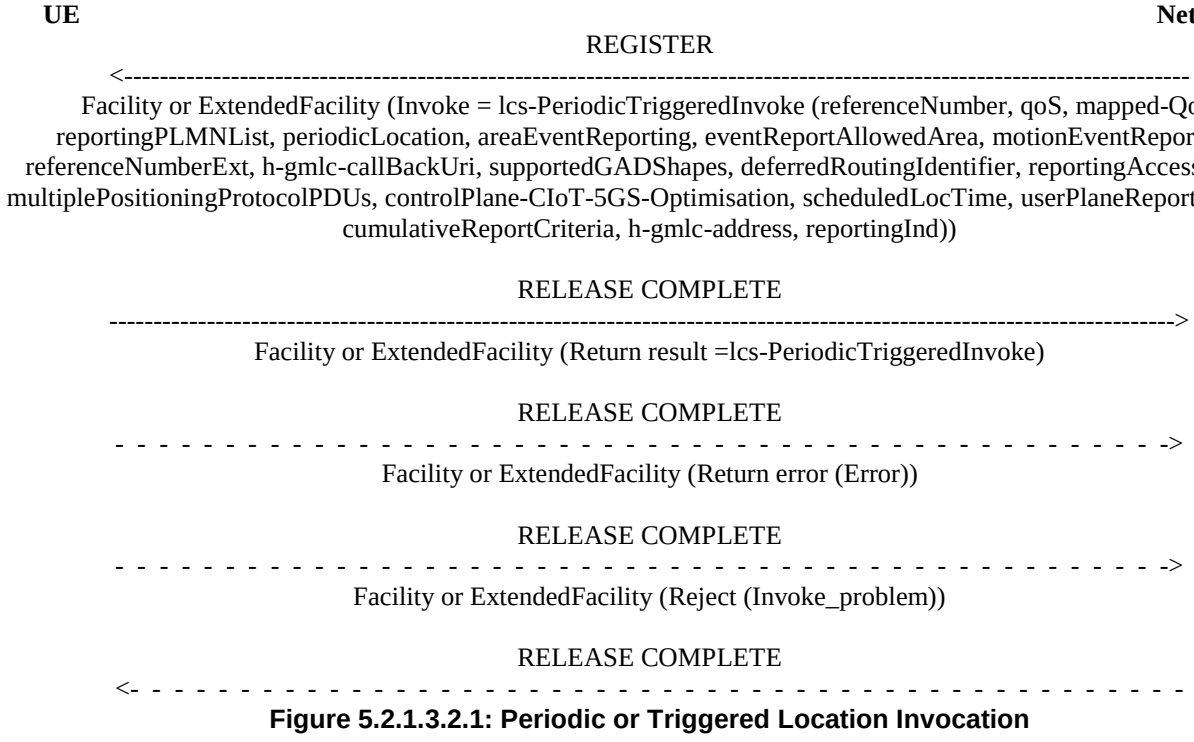


Figure 5.2.1.3.2.1: Periodic or Triggered Location Invocation

NOTE 1: The mappedLcs-QoS is obtained by the UE in 5GS and used for location service continuity from 5GS to EPS for periodic or triggered location event reporting.

NOTE 2: If referenceNumber and h-gmlc-address as defined in 3GPP TS 24.080 [5] are received in the lcs-PeriodicTriggeredInvoke invoke component, they are used for a EPC-(H)GLMC during the procedure of location service continuity between EPS and 5GS.

NOTE 3: If reportingIndication as defined in 3GPP TS 24.080 [5] is received in the lcs-PeriodicTriggeredInvoke invoke component, the reportingIndication is used for determining whether the Event Reporting procedure is triggered when the UE is inside or outside the eventReportAllowedArea.

5.2.1.4 Supplementary Services Cancel Deferred Location

5.2.1.4.1 General

The supplementary services Cancel Deferred Location operation enables the AMF to cancel ongoing periodic or triggered location in a target UE using NAS signalling as described in 3GPP TS 23.273 [2] clause 6.3.3. The lcs-CancelDeferredLocation invoke component, lcs-CancelDeferredLocation return result component, related return error component and related reject component are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. Figure 5.2.1.4.1-1 illustrates an example of the NAS signalling transport.

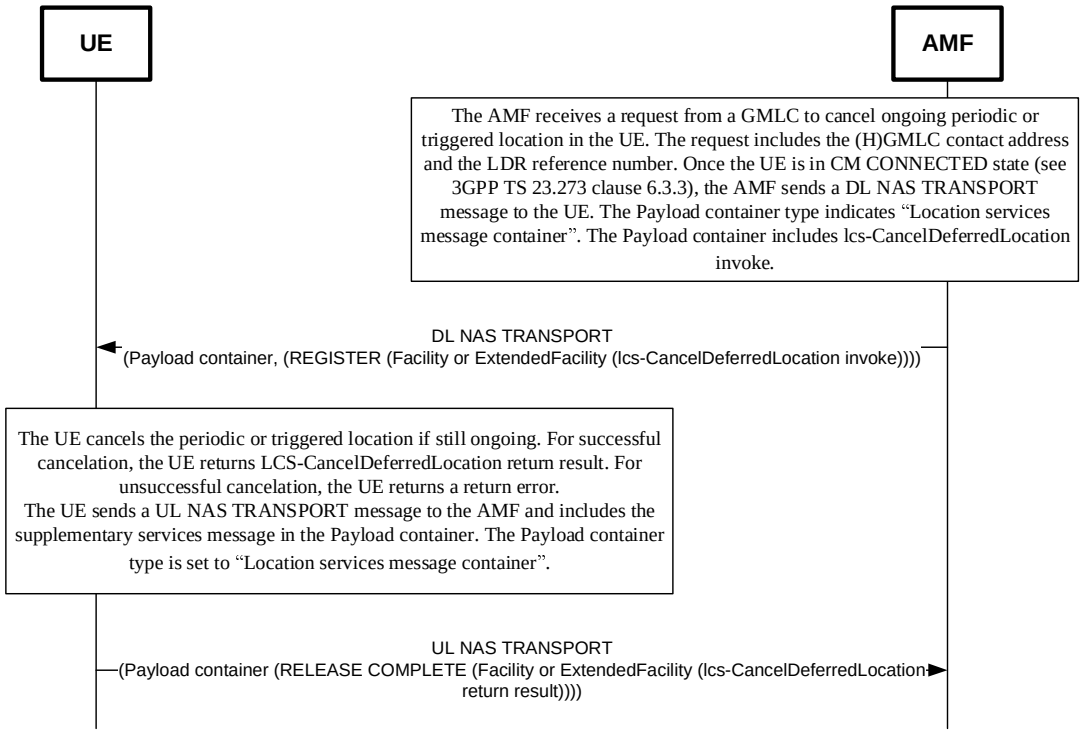


Figure 5.2.1.4.1.1: NAS signalling transport for Cancel Deferred Location

NOTE: The optional Additional information IE of the DL NAS TRANSPORT message or UL NAS TRANSPORT message is not included when the lcs-CancelDeferredLocation invoke component, lcs-CancelDeferredLocation return result component, related return error component or related reject component is transported in the Payload container.

5.2.1.4.2 Normal operation

The AMF invokes a cancel deferred location procedure by sending a REGISTER message containing an lcs-CancelDeferredLocation invoke component to the UE as defined in 3GPP TS 24.080 [5].

The UE shall terminate the ongoing periodic or triggered location if this can be identified from the information in the lcs-CancelDeferredLocation invoke component.

The UE shall then return a RELEASE COMPLETE message containing an lcs-CancelDeferredLocation return result component (see figure 5.2.1.4.2.1).

If the UE is unable to process the request received from the network or cannot identify the ongoing periodic or triggered location to be terminated, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the supplementary services cancel deferred location operation, the network shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the UE. If this timer expires the network shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message.

The use of Facility or ExtendedFacility IE in figure 5.2.1.4.2.1 is specified in clause 5.3.1.

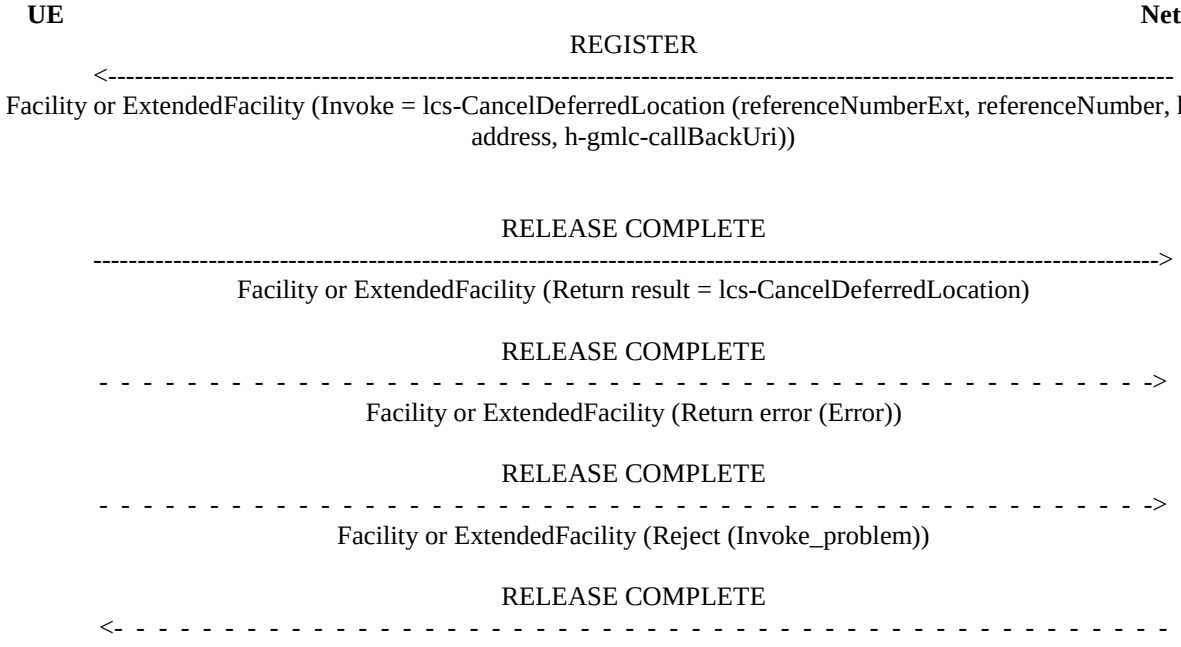


Figure 5.2.1.4.2.1: Cancel Deferred Location

NOTE 1: If the deferred location operation was initiated in 5GS, and the cancel deferred location procedure was triggered in 5GS, then the referenceNumber and the h-gmlc-address are not included in the lcs-CancelDeferredLocation invoke component, and the referenceNumberExt and the h-gmlc-callBackUri are included in the lcs-CancelDeferredLocation invoke component.

NOTE 2: If the UE supports the location service continuity between 5GS and EPS for deferred location as defined in 3GPP TS 24.501 [3], and the deferred location operation was initiated in EPS, then for the cancel deferred location triggered by network, the referenceNumberExt and the h-gmlc-callBackUri are not included in the lcs-CancelDeferredLocation invoke component and the referenceNumber and the h-gmlc-address are included in the lcs-CancelDeferredLocation invoke component.

5.2.1.5 Network initiated PRU disassociation procedure

5.2.1.5.1 General

The supplementary services PRU disassociation operation enables the LMF to disassociate the associated PRU by using NAS signalling as described in clause 6.17.2 of 3GPP TS 23.273 [2]. The NAS signaling are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. The LMF may invoke this procedure prior to the event of becoming unavailable LMF (e.g., for maintenance, removal or replacement of the LMF).

Figure 5.2.1.5.1-1 illustrates an example of the NAS signaling transport for a PRU disassociation procedure.

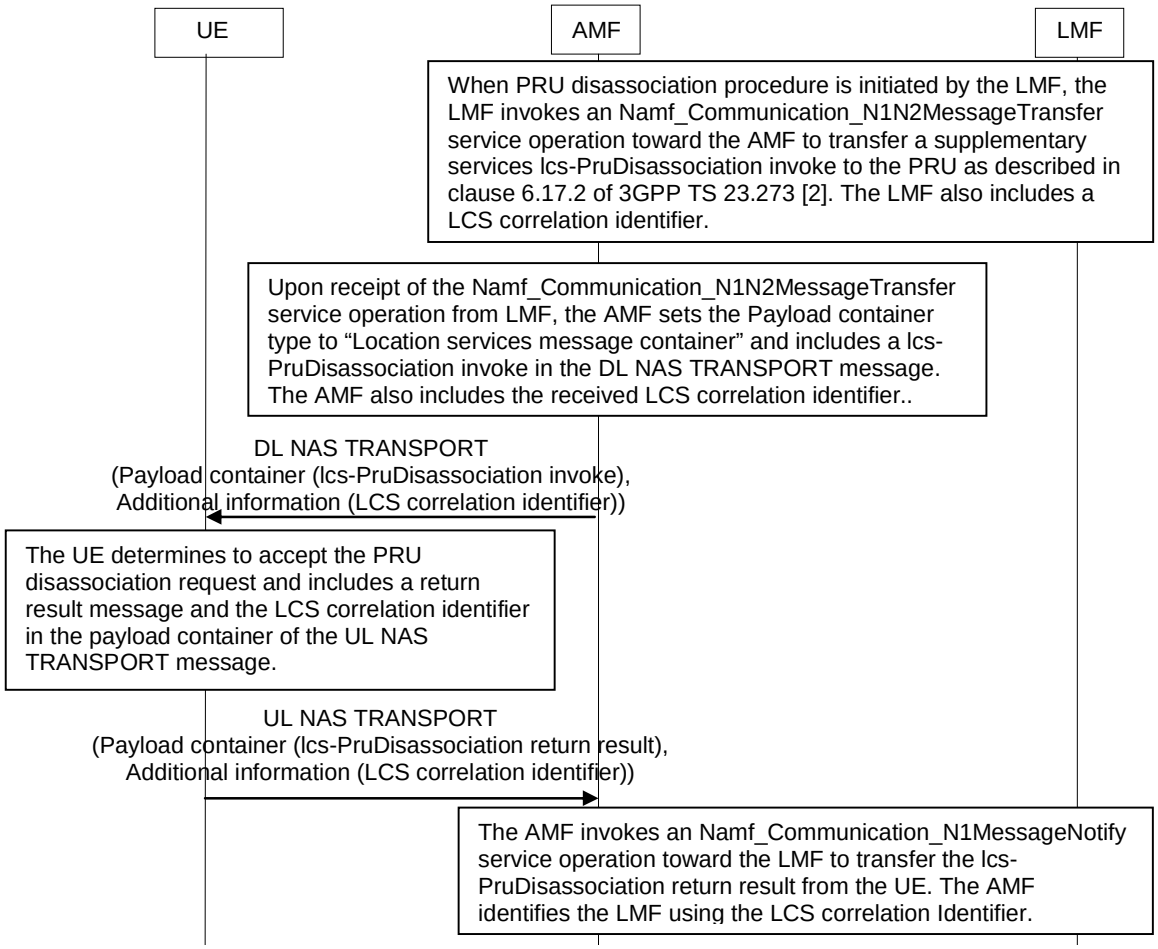


Figure 5.2.1.5.1-1: NAS signalling transport for network initiated PRU disassociation procedure

5.2.1.5.2 Normal operation

The LMF invokes a PRU disassociation procedure by sending lcs-PruDisassociation invoke component to the associated UE operating as PRU as defined in 3GPP TS 24.080 [5].

The UE operating as PRU shall terminate the PRU association with the serving PLMN if this can be identified from the information in the lcs-PruDisassociation invoke component. The UE shall then return a RELEASE COMPLETE message containing a lcs-PruDisassociation return result component (see figure 5.2.1.5.2.1). If the UE operating as PRU receives a new routing ID for a new serving LMF in the lcs-PruDisassociation invoke component from the LMF, the UE operating as PRU may perform a PRU association procedure with the new serving LMF as described in clause 5.2.2.7.

The use of Facility or ExtendedFacility IE in figure 5.2.1.5.2.1 is specified in clause 5.3.1.

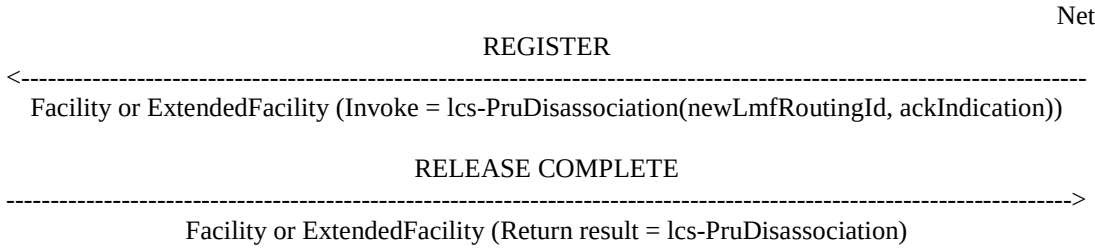


Figure 5.2.1.5.2.1: Network initiated PRU disassociation procedure

NOTE 1: The following IE defined in lcs-PruDisassociation operation in 3GPP TS 24.080 [5] is used for NG-RAN LCS:

- newLmfRoutingId

5.2.1.6 Sidelink Positioning Information Transport

The AMF sends an SLPP message and a Routing identifier which is the associated LCS correlation identifier in the DL NAS TRANSPORT message (refer to 3GPP TS 24.501 [3], 3GPP TS 23.586 [10], 3GPP TS 23.273 [2] clause 6.20). Figure 5.2.1.6 illustrates an example of the NAS signalling transport for downlink SLPP messages.

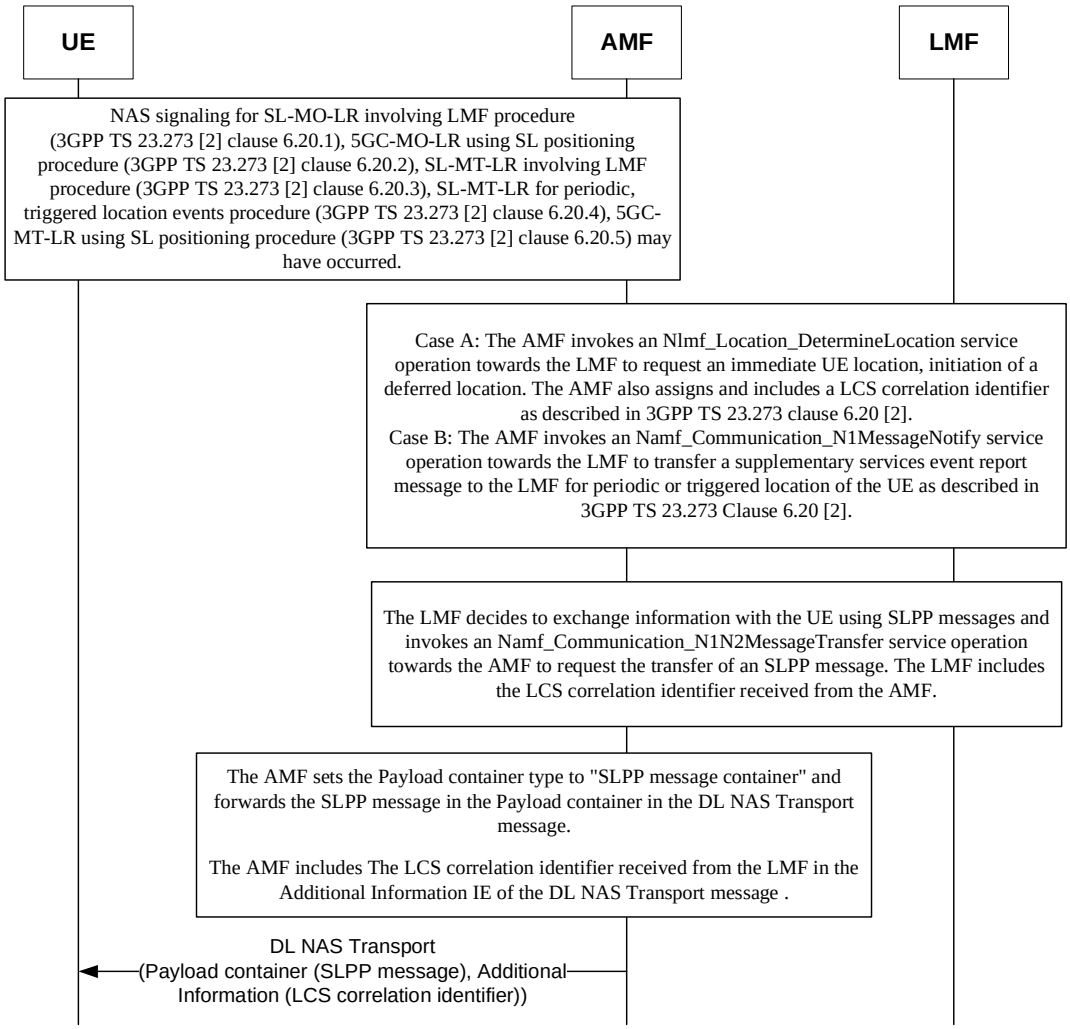


Figure 5.2.1.6.1: NAS signalling transport for downlink SLPP messages

5.2.1.7 Network initiated RSPP supplementary information transport

5.2.1.7.1 General

The network initiated RSPP supplementary information transport procedure enables the LMF to transfer the RSPP supplementary information to the target UE for capability exchange, assistance data exchange or location information exchange as described in clause 6.20.1, clause 6.20.2, clause 6.20.3, clause 6.20.4 or clause 6.20.5 of 3GPP TS 23.273 [2].

The LCS message carrying the lcs-DLRSPPTransport invoke component is transferred from the LMF to the receiving UE via the serving AMF in a DL NAS TRANSPORT message. An acknowledgement message from the receiving UE is returned to the LMF via the serving AMF and is transferred to the AMF in an UL NAS TRANSPORT message.

5.2.1.7.2 Normal operation

In order to initiate the network initiated RSPP supplementary information transport procedure, the LMF shall send a REGISTER message to the UE containing the lcs-DLRSPPTransport invoke component as defined in 3GPP TS 24.080 [5].

In the lcs-DLRSPPTransport invoke component included in the REGISTER message, the LMF shall include rangingSLPPList IE. In the rangingSLPPList, the LMF:

a) may include the embedded SLPP message as defined in 3GPP TS 38.355 [4a] for the receiving UE;

b) shall include the embedded SLPP message(s) as defined in 3GPP TS 38.355 [4a] for the related SL reference UE(s) or the related located UE(s) with the associated application layer ID(s);

NOTE: The SLPP message(s) sent by the LMF include the SLPP message(s) to request capability, to provide assistance data or to request location information as defined in 3GPP TS 38.355 [4a].

c) may include the indication of network assisted sidelink positioning if LMF determines to apply UE based sidelink positioning;

d) may include the scheduled location time; and

e) may include the relatedUEInfo IE indicating information of candidate located UE(s) when the required location result is absolute location.

The receiving UE shall then return a RELEASE COMPLETE message containing a lcs-DLRSPPTransport return result component (see figure 5.2.1.7.2-1) if the receiving UE accepts the DLRSPPTransport invoke component.

If the receiving UE does not accept the DLRSPPTransport invoke component, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the network initiated RSPP supplementary information transport operation, the network shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the UE. If this timer expires the network shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message.

Figure 5.2.1.7.2.1 illustrates the signalling for normal operation between the UE and the network. The use of Facility or ExtendedFacility IE is specified in clause 5.3.1.

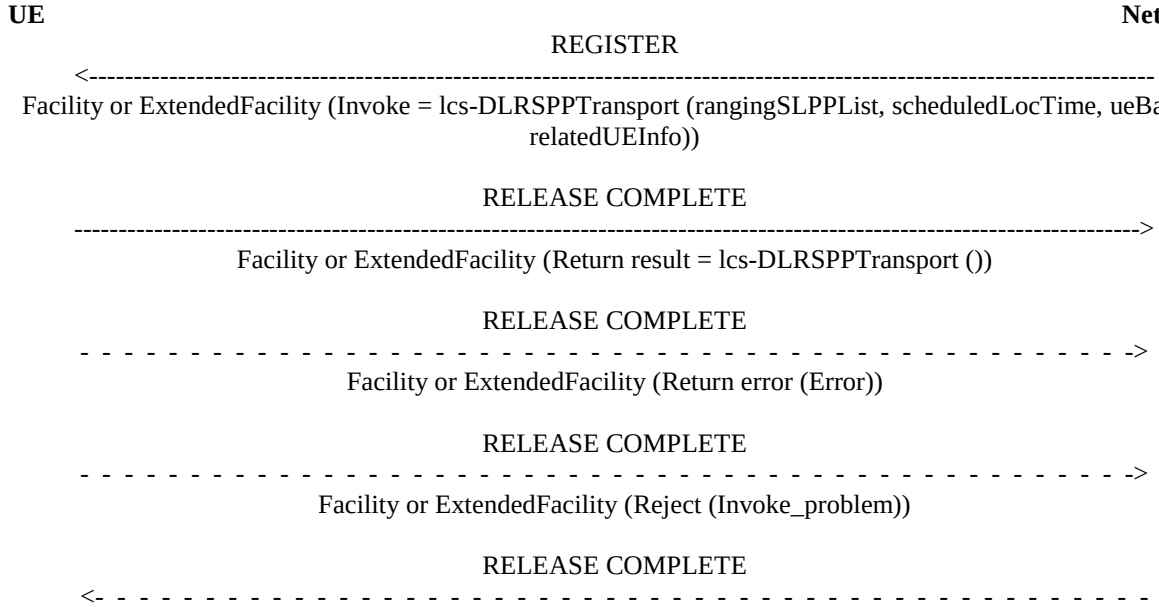


Figure 5.2.1.7.2.1: Network initiated RSPP supplementary information transport procedure

NOTE 1: Only the following IEs defined in DLRSPPTransport operation in 3GPP TS 24.080 [5] are used for ranging and sidelink positioning:

- rangingSLPPList
- scheduledLocTime
- ueBased
- relatedUEInfo

NOTE 2: The rangingSLPPList IE is added to DLRSPPTransport to allow for passing multiple SLPP messages for the UE connecting with the LMF and/or the other related UE. Its ASN.1 description is given in 3GPP TS 24.080 [5].

5.2.1.8 Sidelink mobile terminating location request

5.2.1.8.1 General

The sidelink mobile terminating location request procedure enables the LMF to obtain location information as described in clause 6.20.2, clause 6.20.3, clause 6.20.4 or clause 6.20.5 of 3GPP TS 23.273 [2].

The LCS message carrying the lcs-SLMTLR invoke component is transferred from the LMF to the target UE via the serving AMF in a DL NAS TRANSPORT message. A response from the target UE is returned to the LMF via the serving AMF and is transferred to the AMF in an UL NAS TRANSPORT message.

5.2.1.8.2 Normal operation

In order to initiate the sidelink mobile terminating location request procedure, the LMF shall send a REGISTER message to the UE containing the lcs-SLMTLR invoke component as defined in 3GPP TS 24.080 [5] as described in clause 6.20.2, clause 6.20.3, clause 6.20.4 or clause 6.20.5 of 3GPP TS 23.273 [2]. In the lcs-SLMTLR invoke component, the LMF:

- shall include the relatedUEInfo IE indicating information of candidate located UE(s) when the required location result is absolute location or indicating information of candidate SL reference UE(s) when the required location result is relative location;
- shall include the locatedUEselect IE indicating whether the target UE or the LMF is to select the located UE(s) when the required location result is absolute location; and

- may include the coordinateID IE indicating local coordinate(s) when the required location result is absolute location. The coordinateID IE is used to determine the target UE location in local coordinates.

When the lcs-SLMTLR invoke component is applied for periodic-triggered location events as described in clause 6.20.4 of 3GPP TS 23.273 [2], the LMF additionally:

- shall include the requestedRangingResult IE indicating information of the requested location results (e.g. absolute locations, relative locations, distances, directions, velocities and relative velocities) for the target UE;
- shall include the h-gmlc-address IE indicating the address for the (H)GMLC;
- shall include the referenceNumber IE indicating the LDR reference number;
- shall include periodicLocation IE, areaEventReporting IE, or motionEventReporting IE indicating the reporting criteria;
- shall include the deferredRoutingIdentifier IE indicating identification of a new LMF, if any,which is to serve the UE; and
- may include the lcs-QoS IE indicating the QoS for the requested location results.

If the UE accepts the lcs-SLMTLR invoke component, the UE shall send a RELEASE COMPLETE message to the LMF containing the lcs-SLMTLR return result component. In the lcs-SLMTLR return result component, the UE:

- shall include the relatedUEInfo IE containing information of the related UEs for the ranging and sidelink positioning discovered by the UE, if the lcs-SLMTLR invoke component includes the locatedUEselect IE indicating that the LMF is to select the located UE(s);
- shall include the relatedUEInfo IE containing information of the related UEs for the ranging and sidelink positioning selected by the UE, if the lcs-SLMTLR invoke component includes the locatedUEselect IE indicating that target UE is to select the located UE(s); and
- may include the rangingSLPPList IE containing the embedded SLPP message providing sidelink positioning capabilities of the related UEs as defined in 3GPP TS 38.355 [4a].

If the UE is unable to process or support the request received from the network, the UE shall send a RELEASE COMPLETE message containing a return error component or reject component. Error values are specified in 3GPP TS 24.080 [5].

During the sidelink mobile terminating location request operation, the network shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the UE. If this timer expires the network shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message.

Figure 5.2.1.8.2.1 illustrates the signalling for normal operation between the UE and the network. The use of Facility or ExtendedFacility IE is specified in clause 5.3.1.

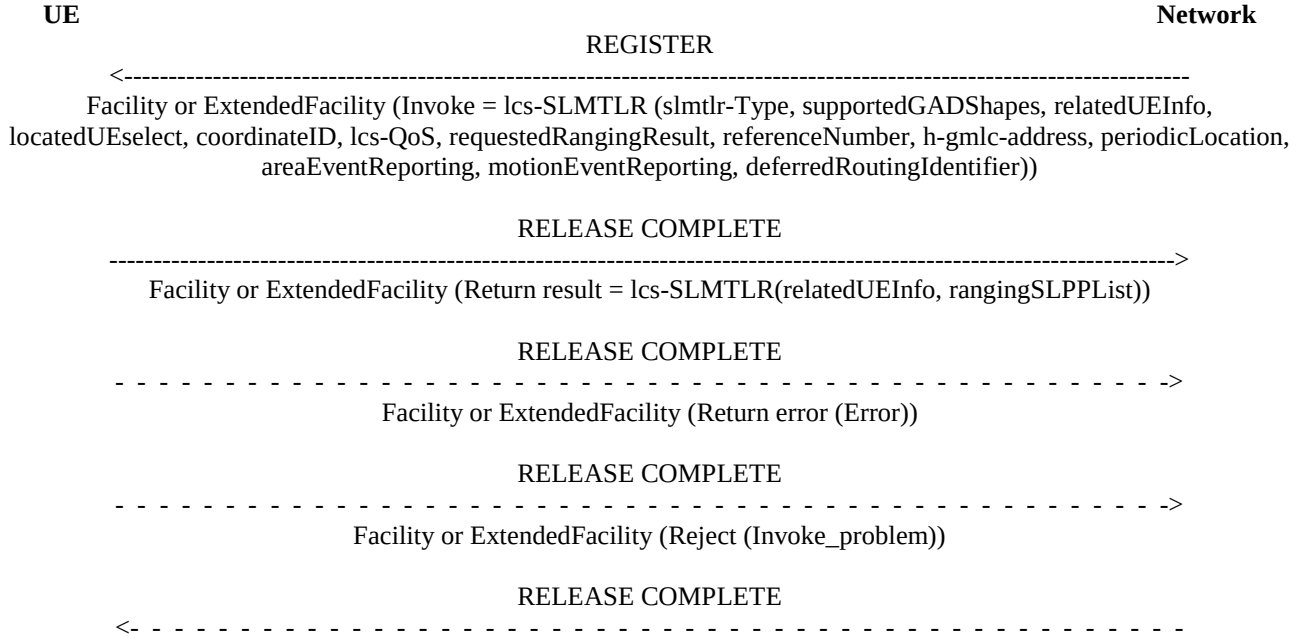


Figure 5.2.1.8.2.1: Sidelink mobile terminating location request procedure

NOTE 1: Only the following IEs defined in lcs-SLMTLR operation in 3GPP TS 24.080 [5] are used for ranging and sidelink positioning:

- slmtlr-Type
- supportedGADShapes
- relatedUEInfo
- locatedUEselect
- coordinateID
- rangingSLPPList
- lcs-QoS
- requestedRangingResult
- referenceNumber
- h-gmlc-address
- periodicLocation
- areaEventReporting
- motionEventReporting
- deferredRoutingIdentifier

NOTE 2: The rangingSLPPList IE is added to the lcs-SLMTLR return result component to allow for passing multiple UE positioning information SLPP messages (e.g. UE capabilities) to the LMF. Its ASN.1 description is given in 3GPP TS 24.080 [5].

5.2.2 Mobile initiated location services operations

5.2.2.1 Mobile Originated Location Request(MO-LR)

5.2.2.1.1 General

The supplementary services MO-LR operation enables the UE to launch MO positioning session or request location assistance data using NAS signaling. The NAS signaling are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. Figure 5.2.2.1.1-1 illustrates an example of the NAS signaling transport for an MO-LR session.

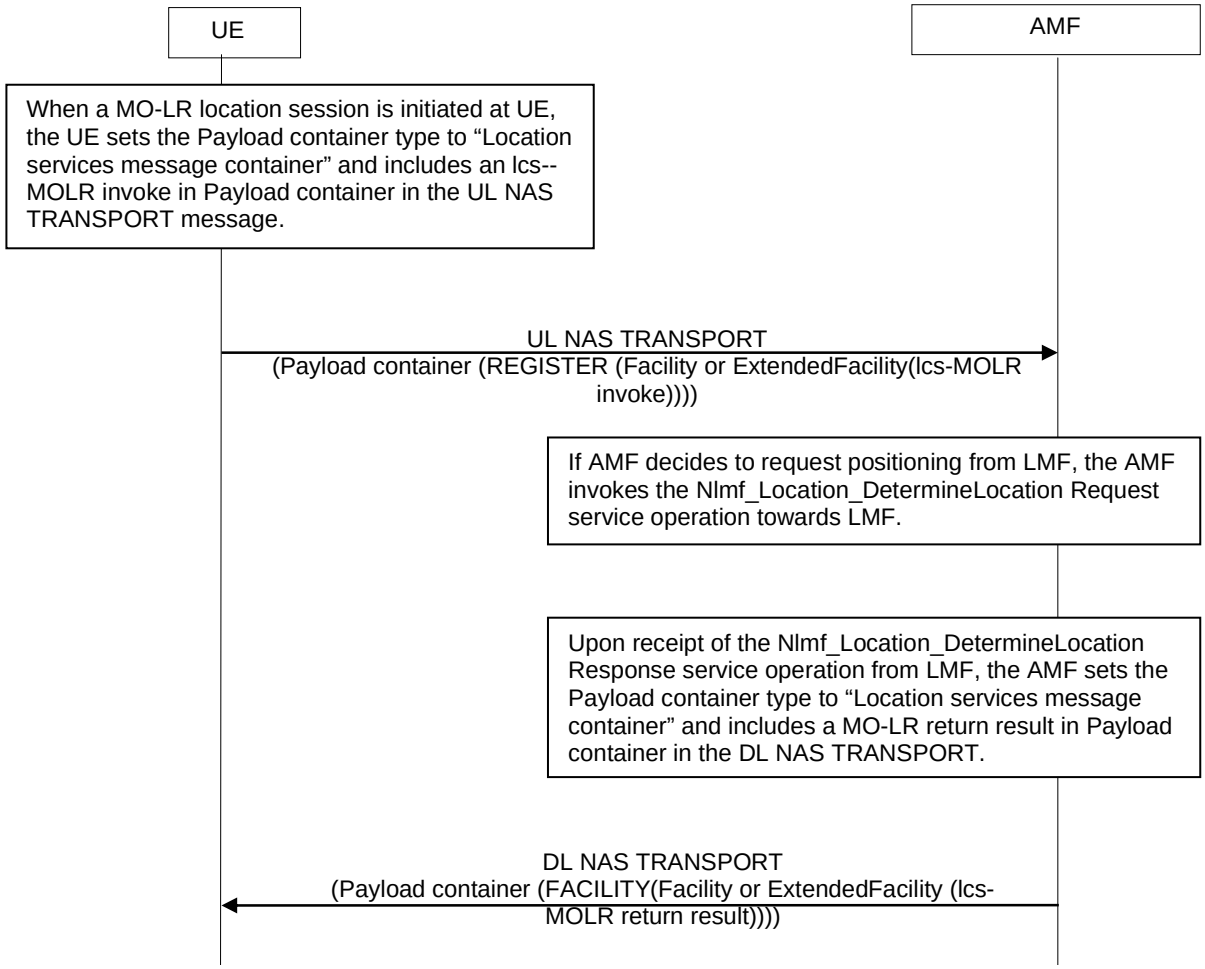


Figure 5.2.2.1.1-1: NAS signaling transport for MO-LR

NOTE: The optional Additional information IE of the UL NAS TRANSPORT message or DL NAS TRANSPORT message is not included when the lcs-MOLR invoke component, lcs-MOLR return result component, related return error component or related reject component is transported in the Payload container.

5.2.2.1.2 Normal operation

The UE invokes an MO-LR by sending a REGISTER message to the network containing an lcs-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

The receiving network entity shall initiate the handling of location request in the network. The network shall pass the result of the location procedure to the UE by sending a FACILITY message to the UE containing an lcs-MOLR return result component. When location estimate is kept in the network entity and this information satisfies the requested accuracy and the requested maximum age of location, then the network may reuse this information and the positioning measurement procedure may be skipped.

The network shall pass the result of the location procedure to the UE only if the location estimate is given in a format that the UE supports, as indicated by either the presence (and content) or the absence of the parameter supportedGADShapes, which may be sent by the UE in the lcs-MOLR invoke component.

The UE may terminate the dialogue by sending a RELEASE COMPLETE message in the case of single location request (see figure 5.2.2.1.1-1). The UE may also initiate another location request operation by sending a FACILITY message to the network containing an lcs-MOLR invoke component (see figure 5.2.2.1.1-2). After the last location request operation the UE shall terminate the dialogue by sending a RELEASE COMPLETE message.

If the network is unable to successfully fulfil the request received from the UE (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5]. If the network is unable to provide a location estimate due to lack of support in the UE for the type of shape of the location estimate, then it shall use the error Facility Not Supported.

If the network has returned a result to the UE in a FACILITY message but, after some PLMN administered time period has elapsed, has not received either a new location request operation in a FACILITY message or a RELEASE COMPLETE message from the UE, the network may clear the transaction by sending a RELEASE COMPLETE message.

During the MO-LR operation the UE shall run a timer T(LCSL). This timer is started when the operation is sent, and stopped when a response is received from the network. If this timer expires the UE shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message, and shall inform the user of the failure.

The use of Facility or ExtendedFacility IE in figure 5.2.2.1.1-1 and figure 5.2.2.1.1-2 are specified in clause 5.3.1.

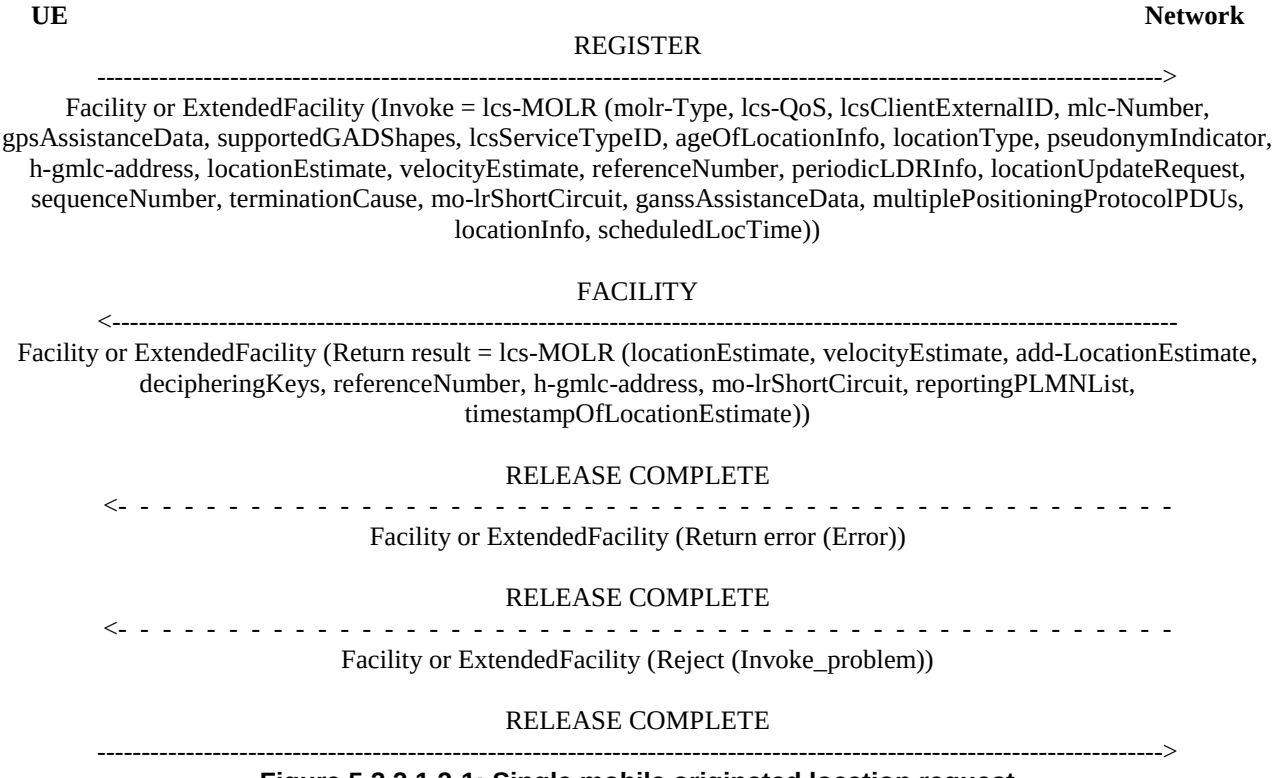


Figure 5.2.2.1.2-1: Single mobile originated location request



Figure 5.2.2.1.2-2: Multiple mobile originated location requests

NOTE 1: Only the following IEs defined in lcs-MOLR operation in 3GPP TS 24.080 [5] are used for NG-RAN LCS:

- molr-Type
- lcs-QoS
- lcsServiceTypeID
- ageOfLocationInfo
- locationType
- mlc-Number

- lcsClientExternalID
- pseudonymIndicator
- supportedGADShapes
- multiplePositioningProtocolPDUs
- locationEstimate
- add-LocationEstimate
- h-gmlc-address
- decipheringKeys
- scheduledLocTime
- velocityEstimate

NOTE 2: The multiplePositioningProtocolPDUs IE is added to the lcs-MOLR invoke component to allow for passing multiple UE positioning information LPP messages (e.g. UE location measurements or UE capabilities) to the LMF for NG-RAN LCS. Its ASN.1 description is given in 3GPP TS 24.080 [5], where the maximum number of LPP messages is specified. There is one or more types of multiple LPP messages can be encapsulated in one multiplePositioningProtocolPDUs IE.

5.2.2.2 UE initiated Cancel Deferred Location

5.2.2.2.1 General

The supplementary services MSCancelDeferredLocation operation enables the UE to cancel ongoing periodic or triggered location in a target LMF using NAS signalling as described in 3GPP TS 23.273 clause 6.3.2 [2]. The lcs-MSCancelDeferredLocation invoke component, lcs-MSCancelDeferredLocation return result component, related return error component and related reject component are transported using the UL NAS TRANSPORT message and the DL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. The deferred routing identifier in the Additional information IE of the UL NAS TRANSPORT message for the cancellation of periodic or triggered location event reporting can be:

a) an LMF ID; or

b) the deferred routing identifier as specified 3GPP TS 23.003 [12], if the deferred location operation was initiated in EPS and the MSCancelDeferredLocation invoke component is triggered after UE moves from EPS to 5GS.

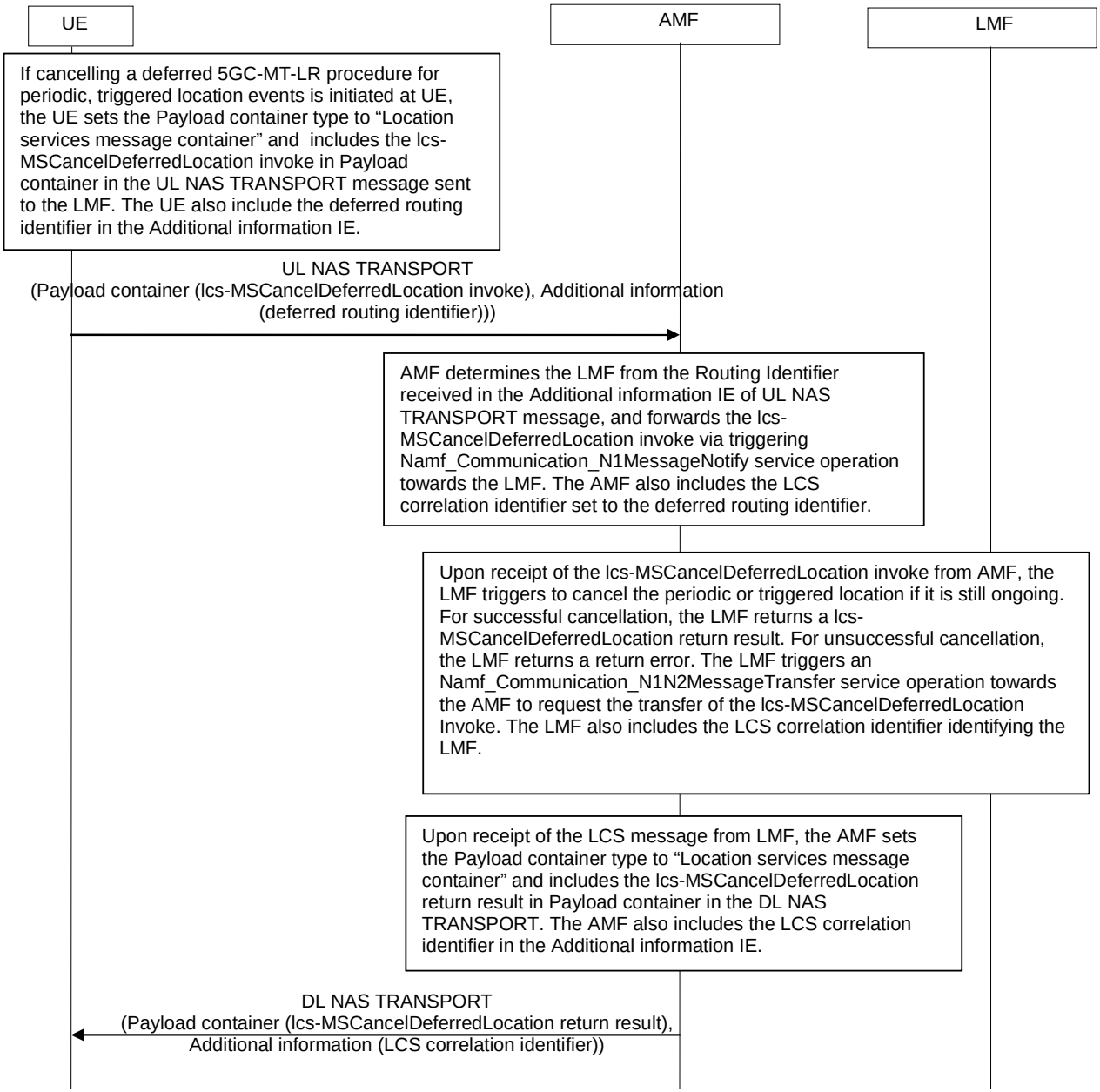


Figure 5.2.2.2.1-1: NAS signalling transport for UE initiated Cancel Deferred Location

5.2.2.2.2 Normal operation

The UE invokes a cancel deferred location procedure by sending a REGISTER message containing an lcs-MSCancelDeferredLocation invoke component to the LMF as defined in 3GPP TS 24.080 [5].

The LMF shall terminate the ongoing periodic or triggered location if this can be identified from the information in the lcs-MSCancelDeferredLocation invoke component.

The LMF shall then return a RELEASE COMPLETE message containing an lcs-MSCancelDeferredLocation return result component (see figure 5.2.2.2.2-1).

NOTE: If the UE has requested the cancellation of the deferred location procedure for the location events reporting over user plane connection and the UE receives the acknowledgment of the cancellation from the LMF, the UE releases the user plane connection to the LCS Client or AF if the release has not been yet initiated.

If the LMF is unable to process the request received from the UE or can not identify the ongoing periodic or triggered location to be terminated, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the UE initiated cancel deferred location operation, the UE shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the network. If this timer expires the UE shall assume that the operation has failed, The UE may terminate the dialogue by sending a RELEASE COMPLETE message, and informs the user of the failure.

The use of Facility or ExtendedFacility IE in figure 5.2.2.2.2-1 is specified in clause 5.3.1.

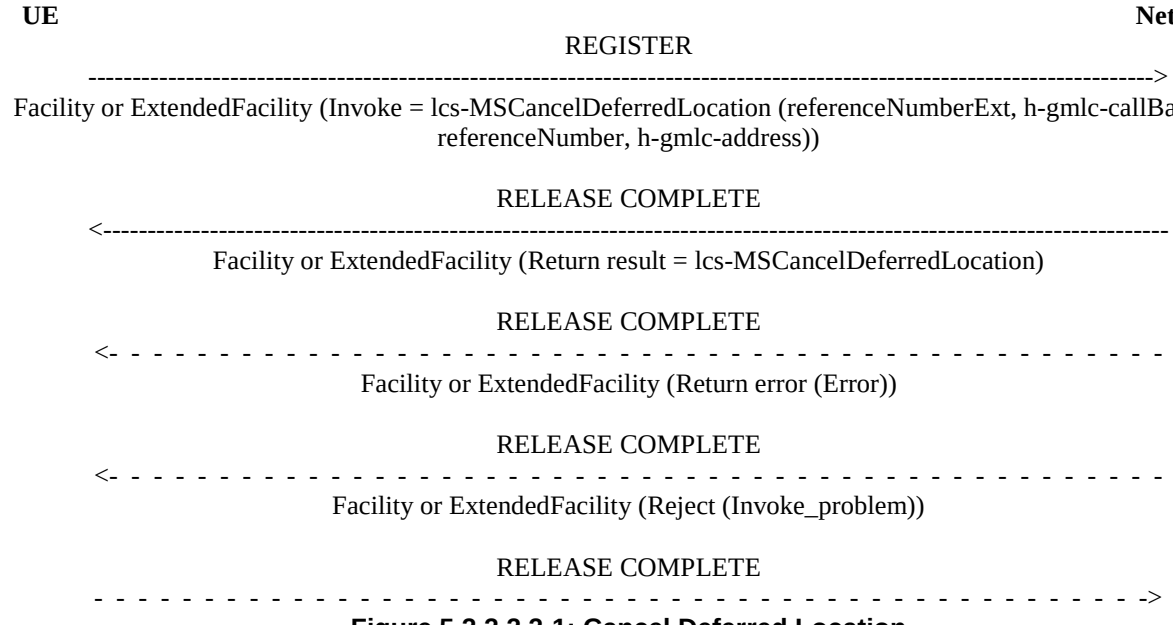


Figure 5.2.2.2.2-1: Cancel Deferred Location

NOTE 1: Only the following IEs defined in lcs-MSCancelDeferredLocation operation in 3GPP TS 24.080 [5] are used for NG-RAN LCS:

- referenceNumberExt
- h-gmlc-callBackUri
- referenceNumber
- h-gmlc-address

NOTE 2: If the deferred location operation was initiated in 5GS, and the cancel deferred location procedure was triggered in 5GS, then the referenceNumber and h-gmlc-address are not included in the lcs-MSCancelDeferredLocation invoke component, and the referenceNumberExt and h-gmlc-callBackUri are included in the lcs-MSCancelDeferredLocation invoke component.

NOTE 3: If the UE supports the location service continuity between 5GS and EPS for deferred location as defined in 3GPP TS 24.501 [3], and the deferred location operation was initiated in EPS, then for the cancel deferred location triggered by UE, the referenceNumberExt and h-gmlc-callBackUri are not included in the lcs-MSCancelDeferredLocation invoke component and the referenceNumber and the h-gmlc-address are included in the lcs-MSCancelDeferredLocation invoke component.

5.2.2.3 UE initiated Positioning Information Transport

The UE sends LPP message and the associated Routing identifier in the UL NAS TRANSPORT message (refer to 3GPP TS 24.501 [3] and 3GPP TS 23.273 [2] clause 6.11.1). Figure 5.2.2.3-1 illustrates an example of the NAS signalling transport for uplink LPP messages.

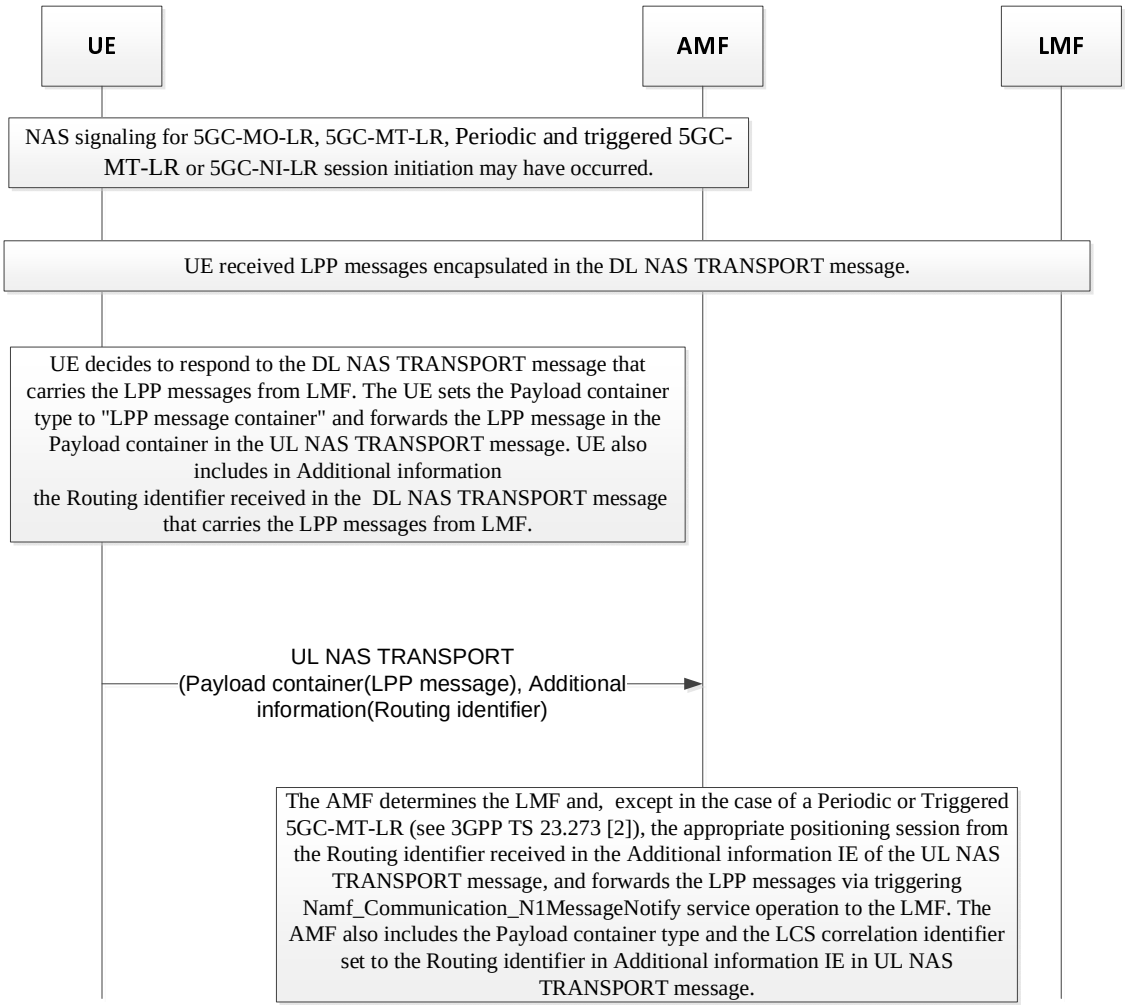


Figure 5.2.2.3-1: NAS signalling transport for uplink LPP messages

5.2.2.4 UE initiated Event Reporting Procedure

5.2.2.4.1 General

The supplementary services EventReport operation enables the UE to report the periodic or triggered location event invoked by the LMF via LCS PeriodicTriggered Invoke operation as described in clause 6.3.1 and clause 6.19.2.2 of 3GPP TS 23.273 [2] or via lcs-SLMTLR invoke operation for periodic, triggered location events as described in clause 6.20.4 of 3GPP TS 23.273 [2] when some certain events are detected in the UE. The UE sends lcs-EventReport invoke component when the UE is inside or outside the eventReportAllowedArea based on the reportingIndication if received in lcs-PeriodicTriggeredInvoke invoke component as described in clause 5.2.1.3. The lcs-EventReport invoke component is transferred to the LMF via the serving AMF in an UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. A response from the LMF may be returned to the UE via the serving AMF and be transferred to the UE in a DL NAS TRANSPORT message. The deferred routing identifier in the Additional information IE of the UL NAS TRANSPORT for reporting the periodic or triggered location event can be:

a) an LMF ID; or

b) the deferred routing identifier as specified 3GPP TS 23.003 [12], if the deferred location operation was initiated in EPS and the lcs-EventReport invoke component is triggered after UE moves from EPS to 5GS.

If the serving LMF is changed, the deferred routing identifier may be included in the lcs-EventReport return result component (refer to clause 6.3.1 of 3GPP TS 23.273 [2]).

The lcs-EventReport invoke component enables the UE to send the cumulative event report over control plane for the periodic or triggered location event with user plane connection invoked by the LMF as described in clause 6.16.1 of 3GPP TS 23.273 [2]. If the cumulative event report timer is expired or the maximum number of events reporting

counter is reached, based on the received value from the LMF, the UE sets the eventType to active status of location reporting over user plane connection and includes the number of location reporting over user plane connection for the cumulative event report in the lcs-EventReport invoke component as defined in 3GPP TS 24.080 [5].

Figure 5.2.2.4.1-1 illustrates an example of the NAS signalling transport for EventReport messages,

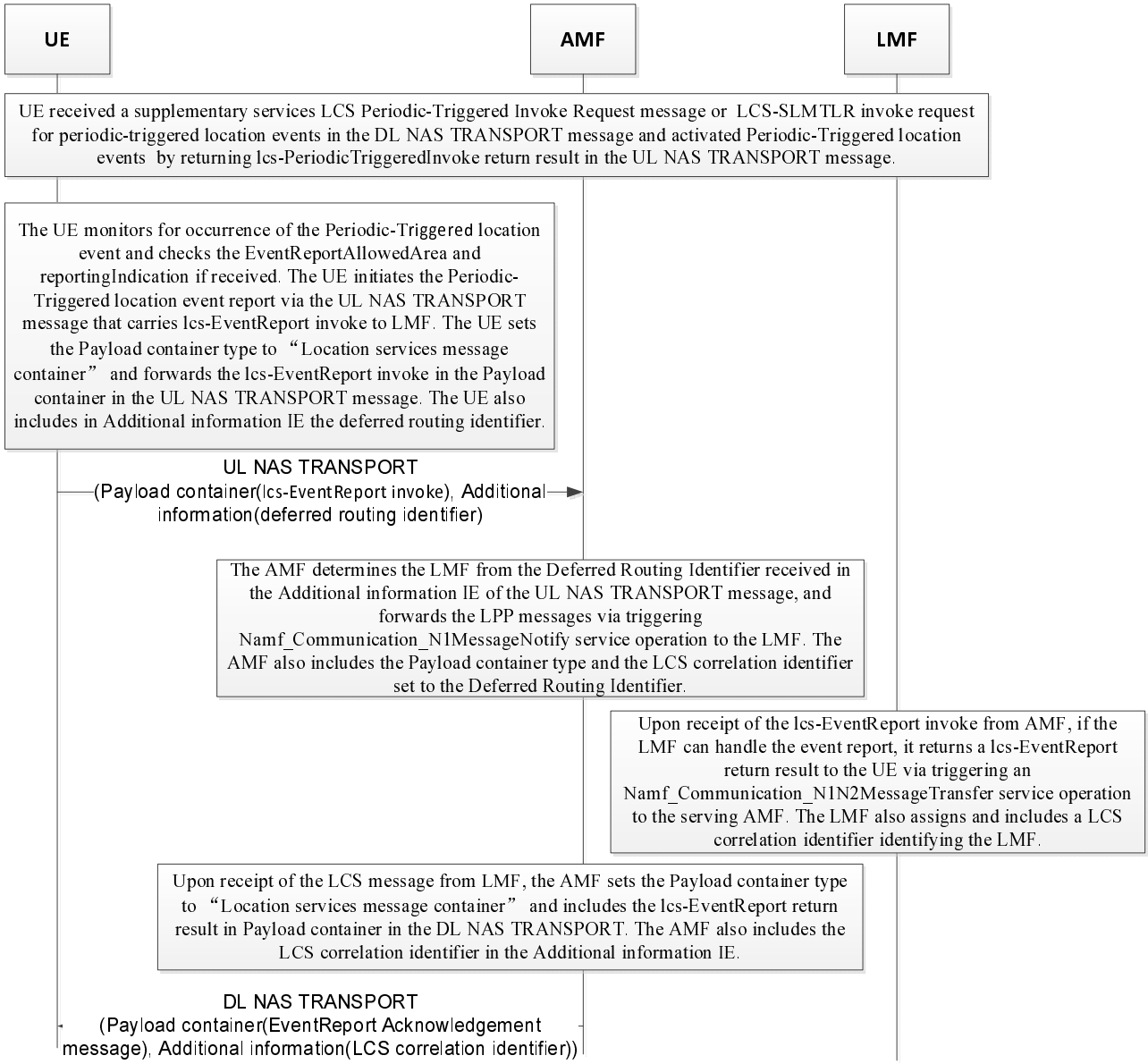


Figure 5.2.2.4.1-1: NAS signalling transport for EventReport messages

5.2.2.4.2 Normal operation

The UE invokes an EventReport procedure by sending a REGISTER message containing an lcs-EventReport invoke component to the LMF as defined in 3GPP TS 24.080 [5].

If the LMF can handle this event report, the LMF shall return a RELEASE COMPLETE message containing an lcs-EventReport return result component (see figure 5.2.2.4.2-1).

If the LMF is unable to process the request received from the UE or cannot identify the ongoing periodic or triggered location to be terminated, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the UE initiated event reporting procedure operation, the UE shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the network. If this timer expires the UE shall assume that the operation has failed. The UE may terminate the dialogue by sending a RELEASE COMPLETE message, and informs the user of the failure.

The use of Facility or ExtendedFacility IE in figure 5.2.2.4.2-1 is specified in clause 5.3.1.

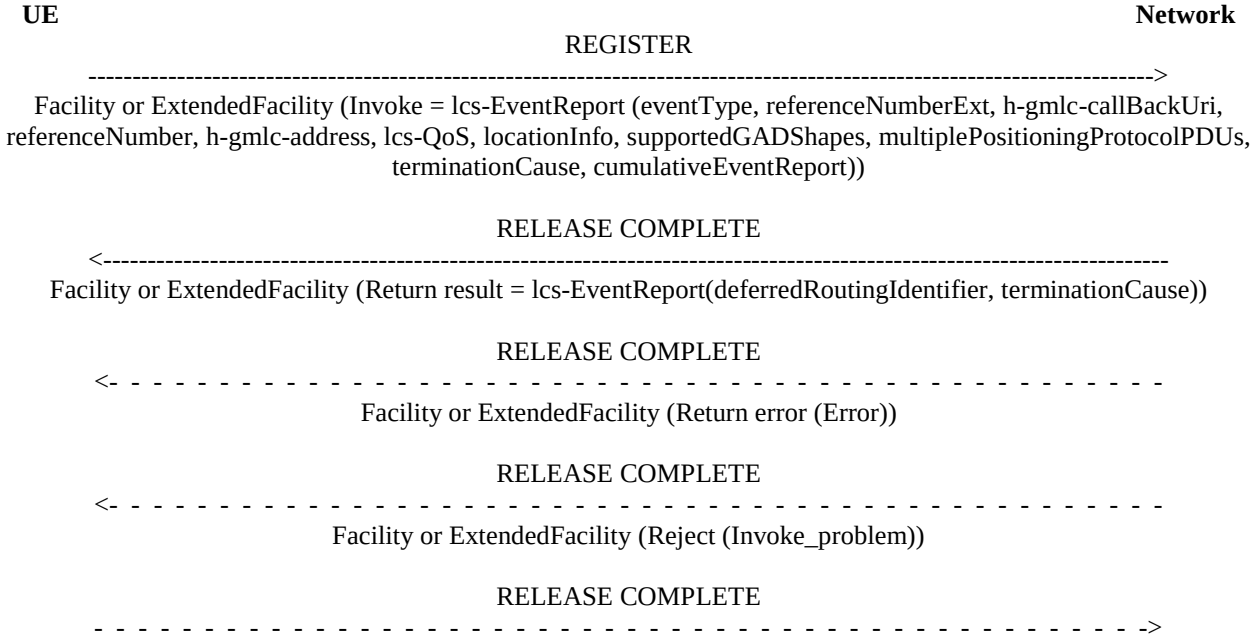


Figure 5.2.2.4.2-1: EventReport

NOTE 1: Only the following identifiers defined in lcs-EventReport operation in 3GPP TS 24.080 [5] are used for NG-RAN LCS:

- eventType
- referenceNumberExt
- h-gmlc-callBackUri
- referenceNumber
- h-gmlc-address
- lcs-QoS
- locationInfo
- supportedGADShapes
- multiplePositioningProtocolPDUs
- terminationCause
- cumulativeEventReport

NOTE 2: If the deferred location operation was initiated in 5GS, and the periodic or triggered location event reporting was triggered in 5GS, then the referenceNumber and the h-gmlc-address are not included in the lcs-EventReport invoke component, and the referenceNumberExt and the h-gmlc-callBackUri are included in the lcs-EventReport invoke component.

NOTE 3: If the UE supports the location service continuity between 5GS and EPS for deferred location as defined in 3GPP TS 24.501 [3], and the deferred location operation was initiated in EPS, then for each periodic or triggered location event report in 5GS, the referenceNumberExt and h-gmlc-callBackUri are not included in the lcs-EventReport invoke component, and the referenceNumber and h-gmlc-address are included in the lcs-EventReport component.

5.2.2.5 UE Location Privacy Setting Procedure

5.2.2.5.1 General

The supplmentary services LocationPrivacySetting operation enables the UE to update the UE Location Privacy Indication information via UE Location Privacy Setting procedure as described in clause 6.12.1 of 3GPP TS 23.273 [2] when the UE has generated or updated the UE Location Privacy Indication. The lcs-LocationPrivacySetting invoke component is transferred to the serving AMF in an UL NAS TRANSPORT message and an acknowledgement from the serving AMF may be returned to the UE in DL NAS TRANSPORT message defined in 3GPP TS 24.501 [3].

Figure 5.2.2.5.1-1 illustrates an example of the NAS signalling transport for UE Location Privacy Setting procedure,

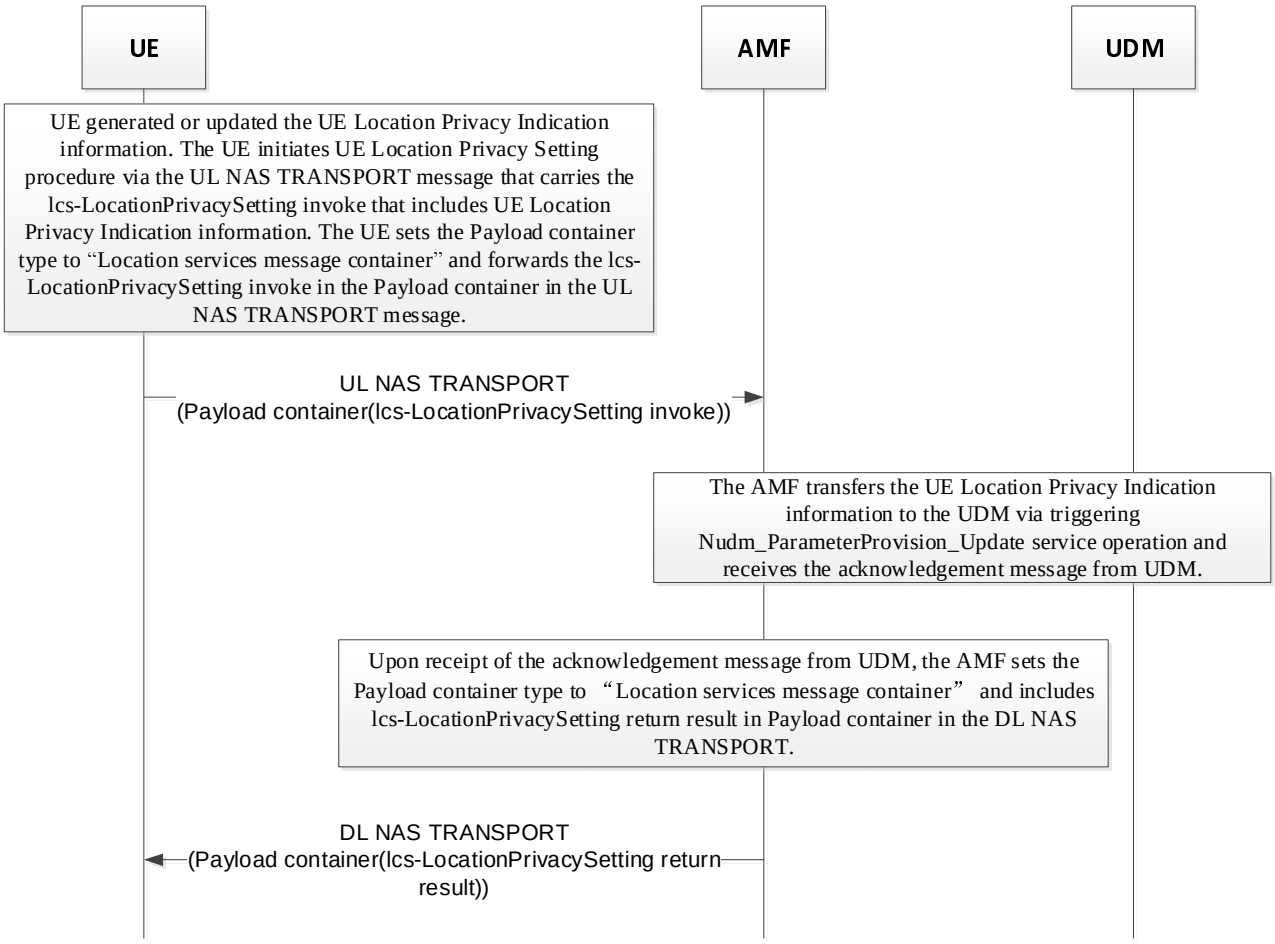


Figure 5.2.2.5.1-1: NAS signalling transport for LocationPrivacySetting messages

5.2.2.5.2 Normal operation

The UE invokes a Location Privacy Setting procedure by sending a REGISTER message containing an lcs-LocationPrivacySetting invoke component as defined in 3GPP TS 24.080 [5]. UE's Location Privacy Indication information is included in lcs-LocationPrivacySetting invoke component.

The AMF shall transfer the UE Location Privacy Indication information in lcs-LocationPrivacySetting invoke component to UDM.

The AMF shall return a RELEASE COMPLETE message containing a LocationPrivacySetting Acknowledgement component (see figure 5.2.2.5.2-1) if the acknowledgement message is received from UDM.

If the AMF is unable to process the lcs-LocationPrivacySetting invoke component received from the UE or cannot identify that provisioning the parameters of UE Location Privacy Setting information to UDM is executed successfully, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the UE initiated location privacy setting procedure operation, the UE shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the network. If this timer expires the UE shall assume that the operation has failed. The UE may terminate the dialogue by sending a RELEASE COMPLETE message, and informs the user of the failure.

The use of Facility or ExtendedFacility IE in figure 5.2.2.5.2-1 is specified in clause 5.3.1.

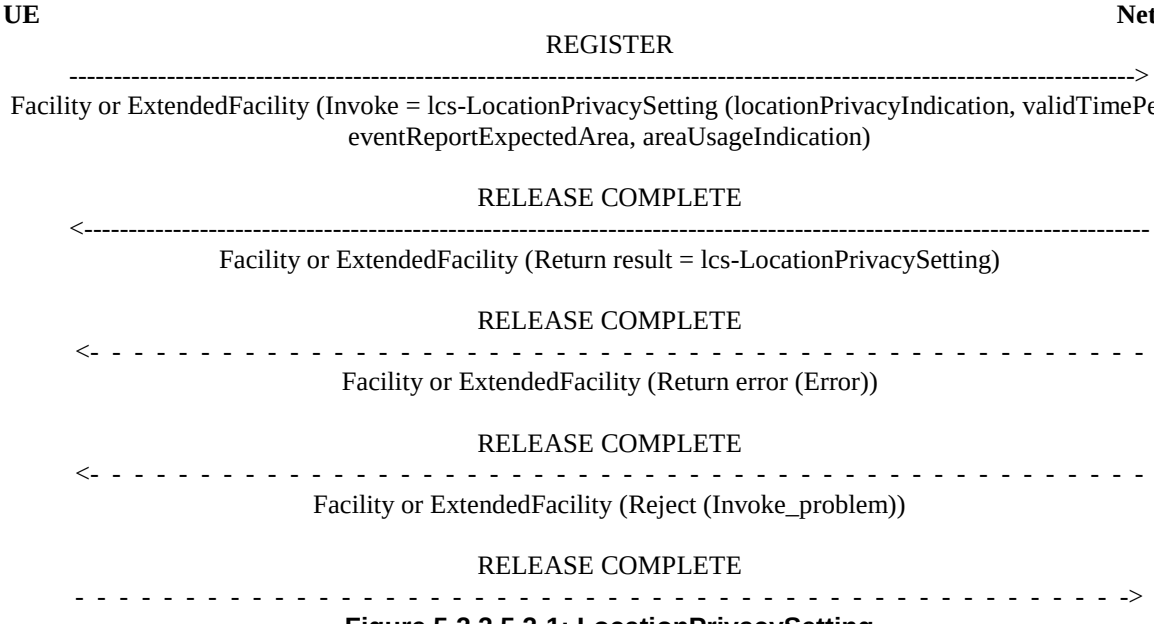


Figure 5.2.2.5.2-1: LocationPrivacySetting

NOTE: Only the following identifiers defined in lcs-LocationPrivacySetting operation in 3GPP TS 24.080 [5] are used for NG-RAN LCS:

- locationPrivacyIndication
- validTimePeriod
- eventReportExpectedArea
- areaUsageIndication

5.2.2.6 UE initiated Event Reporting Procedure for Low Power Event Reporting and Triggered 5GC-MT-LR

5.2.2.6.1 General

The supplementary services EventReport operation enables the UE to report the periodic or triggered location event invoked by the LMF via LCS PeriodicTriggered Invoke operation as described in clause 6.7.1 and in clause 6.19.2.2 of 3GPP TS 23.273 [2] when certain events are detected in the UE and when the UE supports and the LMF allows the use of Control Plane CIoT 5GS Optimisation. The lcs-EventReport invoke component is transferred to the LMF via the serving AMF in a CONTROL PLANE SERVICE REQUEST message defined in 3GPP TS 24.501 [3]. A response from the LMF may be returned to the UE via the serving AMF and be transferred to the UE in a DL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. The deferred routing identifier in the Additional information IE of the CONTROL PLANE SERVICE REQUEST message for reporting the periodic or triggered location event can be:

a) an LMF ID; or

b) the deferred routing identifier as specified 3GPP TS 23.003 [12], if the deferred location operation was initiated in EPS and the lcs-EventReport invoke component is triggered after UE moves from EPS to 5GS.

If the serving LMF is changed, the deferred routing identifier may be included in the EventReport Acknowledgement message (refer to clause 6.7.2 of 3GPP TS 23.273 [2]).

Figure 5.2.2.6.1-1 illustrates an example of the NAS signalling transport for EventReport messages.

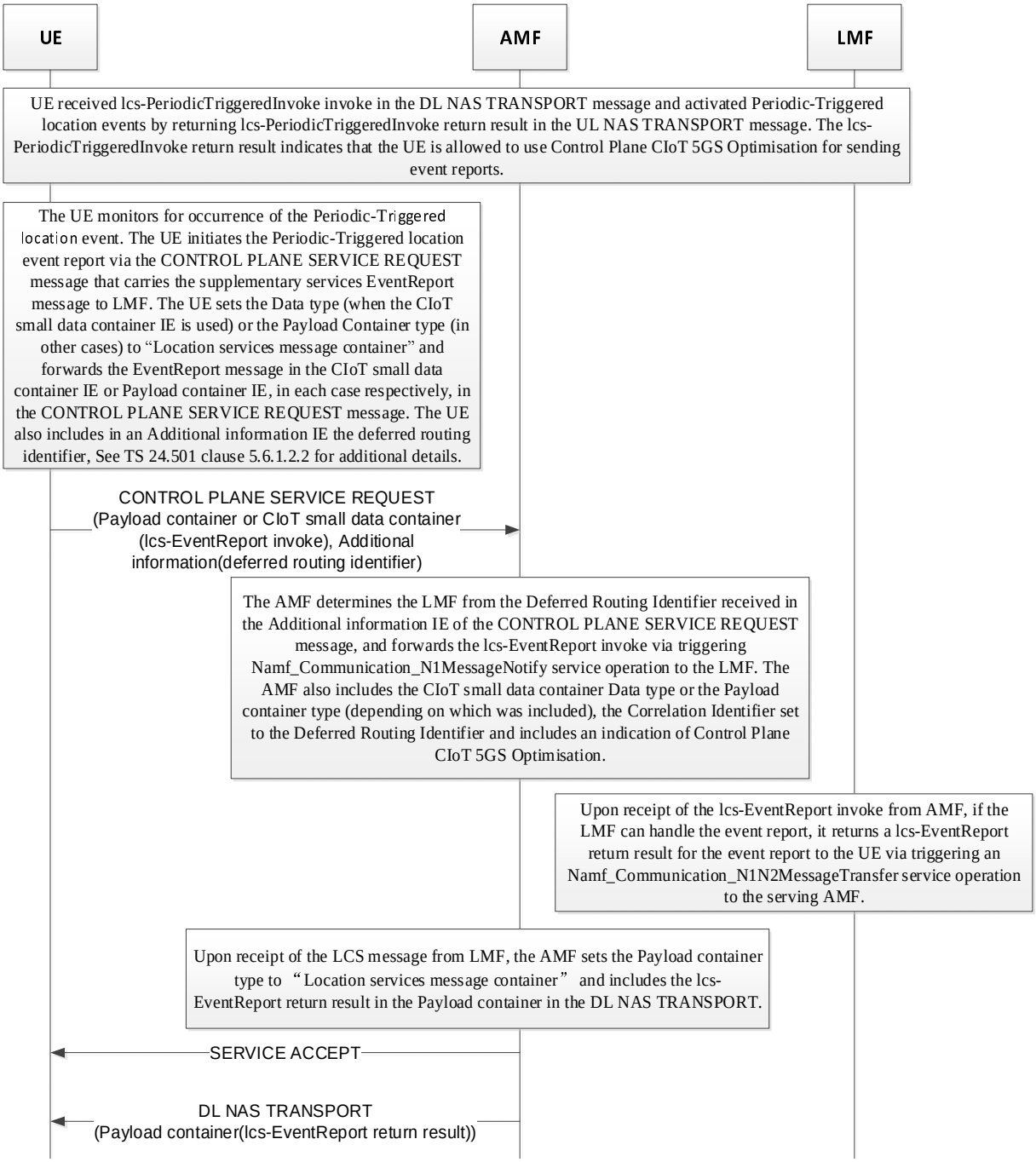


Figure 5.2.2.6.1-1: NAS signalling transport for EventReport messages using Low Power Event Reporting and Triggered 5GC-MT-LR

5.2.2.6.2 Normal operation

Normal operation is as described for the UE initiated Event Reporting Procedure in clause 5.2.2.4.2.

5.2.2.7 PRU association procedure

5.2.2.7.1 General

The supplementary services PRU association operation enables the PRU to associate with a serving LMF by using NAS signalling as described in clause 6.17.1 of 3GPP TS 23.273 [2]. The NAS signaling are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3].

The supplementary services PRU association operation also enables to perform a PRU association update to inform the serving LMF of the continued availability of the PRU or to inform the serving LMF of change to the location of the PRU (e.g. a change of tracking area or change of serving AMF), a change of the PRU positioning capabilities or a change to the PRU ON/OFF state.

Figure 5.2.2.7.1-1 illustrates an example of the NAS signaling transport for a PRU association procedure.

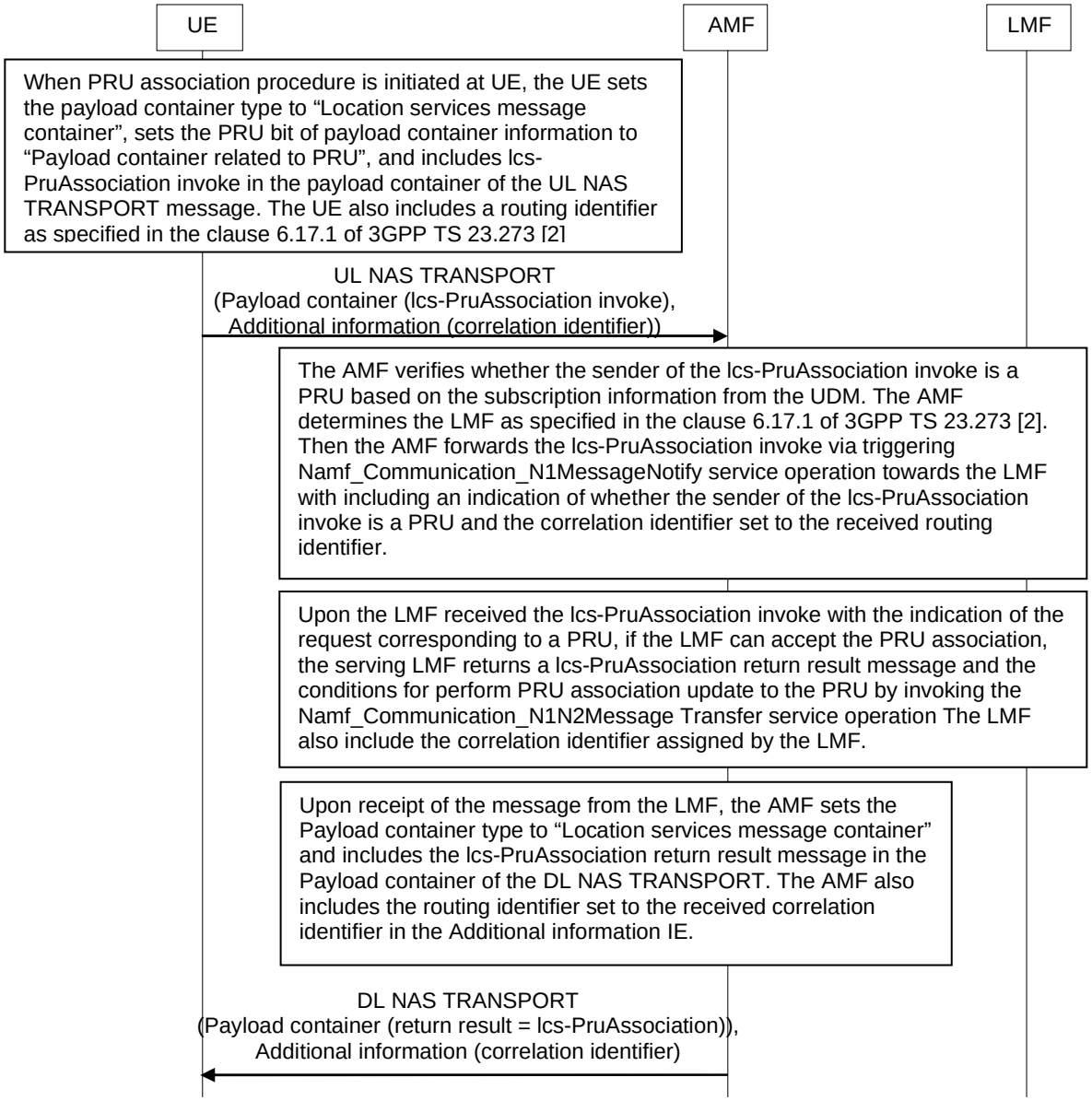


Figure 5.2.2.7.1-1: NAS signalling transport for PRU association procedure

5.2.2.7.2 Normal operation

The UE operating as PRU invokes a PRU association procedure by sending a REGISTER message containing a lcs-PruAssociation invoke component to the serving LMF as defined in 3GPP TS 24.080 [5]. The lcs-PruAssociation invoke component shall include a type for the PRU association request whether it is initial PRU association request or PRU association update, the positioning capabilities of the UE operating as PRU, PRU ON/OFF state, and optionally the location information if available. The UE operating as PRU may include the pre-configured routing ID for an initial association, or the routing ID received from the previous PRU association procedure in the UL NAS TRANSPORT message.

The LMF shall return a RELEASE COMPLETE message containing a lcs-PruAssociation return result component if the LMF accepts the PRU association. The lcs-PruAssociation return result component shall indicate the conditions for performing PRU association updates with the serving LMF. The conditions for performing PRU association update may

include a periodic PRU association update timer and PRU association update based on a change of PRU location, change of PRU TAI, change of serving AMF.

If the LMF is unable to process the lcs-PruAssociation invoke component received from the UE operating as PRU as specified in clause 6.17.1 of 3GPP TS 23.273 [2], the LMF shall return a RELEASE COMPLETE message containing reject component. The reject component may include routing ID of the serving LMF if new LMF is selected to serve the UE operating as PRU.

The use of Facility or ExtendedFacility IE in figure 5.2.2.7.2.1 is specified in clause 5.3.1.

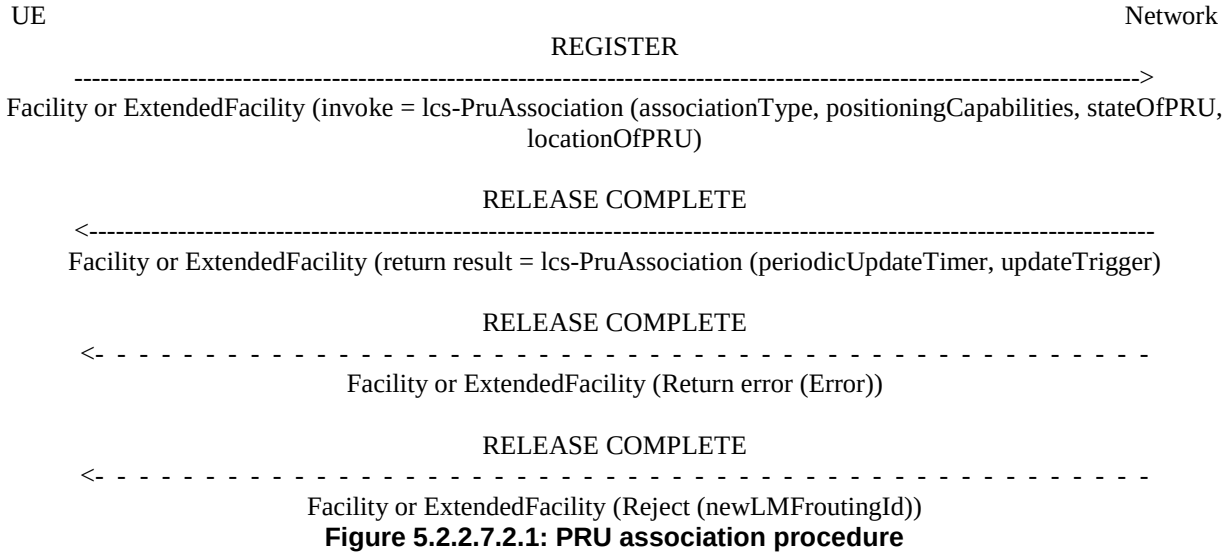


Figure 5.2.2.7.2.1: PRU association procedure

5.2.2.8 UE initiated PRU disassociation procedure

5.2.2.8.1 General

The supplementary services PRU disassociation operation enables the PRU to disassociate from a serving LMF by using NAS signalling as described in clause 6.17.3 of 3GPP TS 23.273 [2]. The NAS signaling are transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. The UE operating as PRU may invoke this procedure prior to the event of becoming unavailable PRU (e.g., for a software upgrade or power down).

Figure 5.2.2.8.1-1 illustrates an example of the NAS signaling transport for a PRU disassociation procedure.

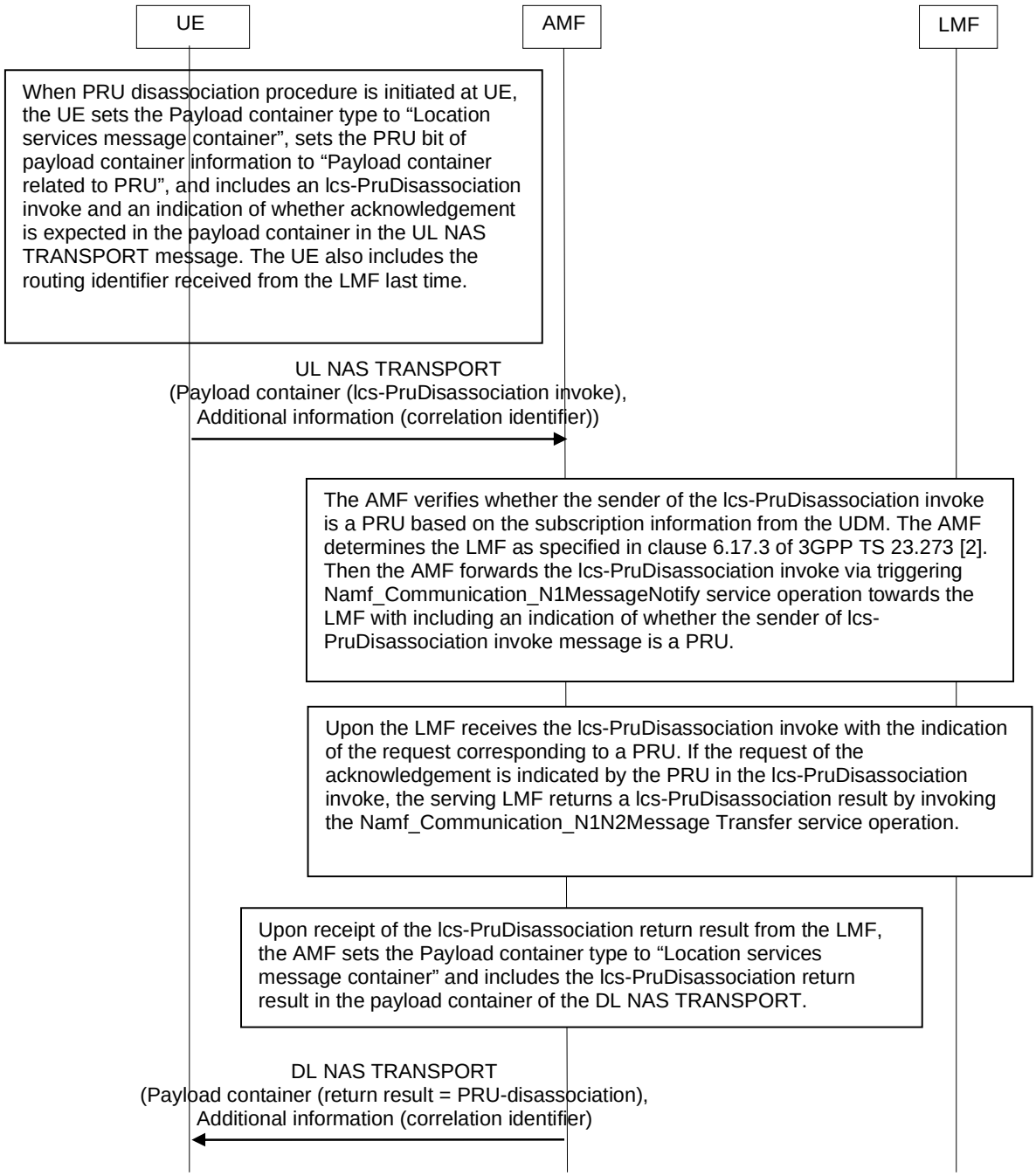


Figure 5.2.2.8.1-1: NAS signalling transport for UE initiated PRU disassociation procedure

5.2.2.8.2 Normal operation

The UE operating as PRU invokes a PRU disassociation procedure by sending lcs-PruDisassociation invoke component to the serving LMF as defined in 3GPP TS 24.080 [5]. The lcs-PruDisassociation invoke component shall include an indication whether an acknowledgement from the LMF is expected or not. The UE operating as PRU shall include the routing ID received from the previous PRU association procedure in the UL NAS TRANSPORT message.

The LMF shall terminate the PRU association with the associated UE operating as PRU if this can be identified from the information in the lcs-PruDisassociation invoke component. The LMF shall then return a RELEASE COMPLETE message containing a lcs-PruDisassociation return result component (see figure 5.2.2.8.2-1). If the UE operating as PRU has not indicated that an acknowledgement is expected, the LMF may disassociate the UE operating as PRU locally without return the RELEASE COMPLETE message.

The use of Facility or ExtendedFacility IE in figure 5.2.2.8.2-1 is specified in clause 5.3.1.

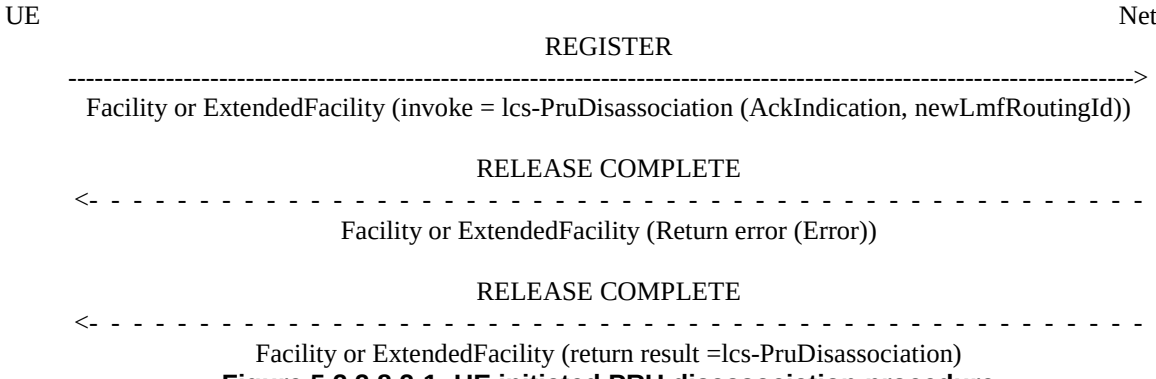


Figure 5.2.2.8.2-1: UE initiated PRU disassociation procedure

5.2.2.9 Sidelink Mobile Originated Location Request (SL-MO-LR)

5.2.2.9.1 General

The supplementary services SL-MO-LR operation enables the target UE to launch the ranging and sidelink MO positioning session or request location assistance data using NAS signaling. The NAS signaling is transported using the DL NAS TRANSPORT message and the UL NAS TRANSPORT message defined in 3GPP TS 24.501 [3]. Figure 5.2.2.9.1-1 illustrates an example of the NAS signaling transport for an SL-MO-LR.

The SL-MO-LR operation can be triggered by a target UE as specified in clause 6.20.1 of 3GPP TS 23.273 [2].

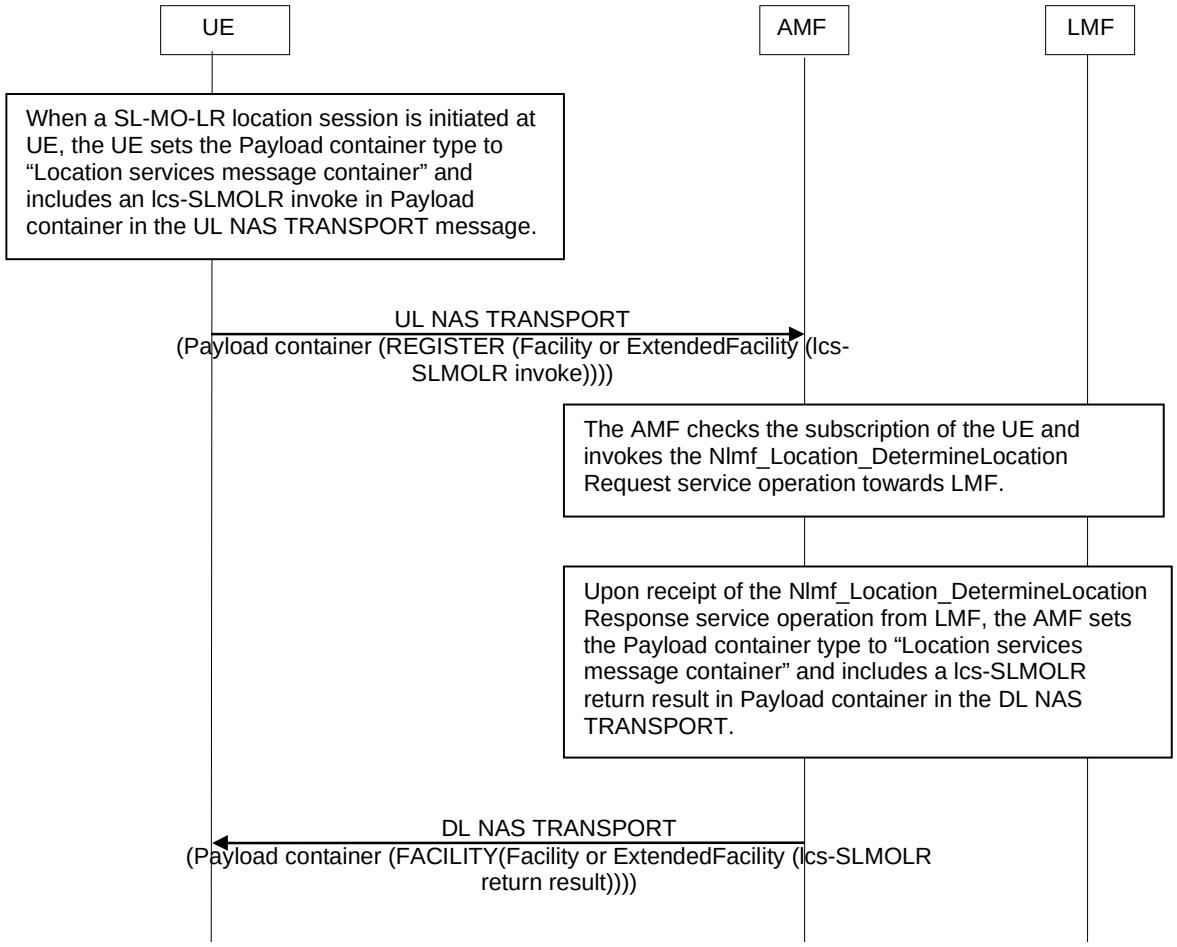


Figure 5.2.2.9.1-1: NAS signalling transport for successful SL-MO-LR

NOTE: The Additional information IE of the UL NAS TRANSPORT message or DL NAS TRANSPORT message is not included when the lcs-SLMOLR invoke component, lcs-SLMOLR return result component, related return error component or related reject component is transported in the Payload container.

5.2.2.9.2 Normal operation

The UE invokes an SL-MO-LR by sending a REGISTER message to the network containing an lcs-SLMOLR invoke component. SS Version Indicator value 1 or above shall be used.

The receiving network entity shall initiate the handling of location request in the network. The network shall pass the result of the location procedure to the UE by sending a FACILITY message to the UE containing an lcs-SLMOLR return result component. When location estimate is kept in the network entity and this information satisfies the requested accuracy and the requested maximum age of location, then the network may reuse this information and the positioning measurement procedure may be skipped.

The network shall pass the result of the location procedure to the UE only if the location estimate is given in a format that the UE supports, as indicated by either the presence (and content) or the absence of the parameter supportedGADShapes, which may be sent by the UE in the lcs-SLMOLR invoke component operation.

The UE may terminate the dialogue by sending a RELEASE COMPLETE message in the case of single location request (see figure 5.2.2.9.2-1). The UE may also initiate another location request operation by sending a FACILITY message to the network containing an lcs-SLMOLR invoke component (see figure 5.2.2.9.2-2). After the last location request operation the UE shall terminate the dialogue by sending a RELEASE COMPLETE message.

When the network is congested and if the network decides not to perform the location procedure, the network may allow UE only ranging and sidelink positioning for a given time duration by sending a RELEASE COMPLETE message to the UE containing an lcs-SLMOLR return result component including a ueOnlyRSLPosAllowed IE. The network shall provide the time duration in the ueOnlyRSLPosAllowed IE.

Upon receiving the lcs-SLMOLR return result component including a ueOnlyRSLPosAllowed IE, the UE may perform the UE only sidelink positioning procedure as specified in clause 6.6 of 3GPP TS 23.586 [10] within the time duration provided in the ueOnlyRSLPosAllowed IE.

If the network is unable to successfully fulfil the request received from the UE (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5]. If the network is unable to provide a location estimate due to lack of support in the UE for the type of shape of the location estimate, then it shall use the error Facility Not Supported.

If the network has returned a result to the UE in a FACILITY message but, after some PLMN administered time period has elapsed, has not received either a new location request operation in a FACILITY message or a RELEASE COMPLETE message from the UE, the network may clear the transaction by sending a RELEASE COMPLETE message.

During the lcs-SLMOLR operation, the UE shall run a timer T5101. This timer is started when the lcs-SLMOLR invoke component is sent, and stopped when a lcs-SLMOLR return result component, return error component or reject component is received from the network. If this timer expires the UE shall assume that the operation has failed, and may terminate the dialogue by sending a RELEASE COMPLETE message, and shall inform the user of the failure.

In the lcs-SLMOLR invoke component included in the REGISTER or FACILITY message, the UE:

a) shall include the slmolr-Type IE set to rangingSidelink;

b) shall include the relatedUEInfo IE containing one entry for each of the one or more SL reference UEs or located UEs, containing application layer ID of the SL reference UE and the slReferenceUE role or application layer ID of the located UE and the locatedUE role;

c) shall include the calculationAssistIndicator IE, if the UE needs the location calculation assistance;

d) shall include the preferredRangingResult IE set to absoluteLocationIndicator, absoluteVelocityIndicator, relativeLocationIndicator, rangeDirection, relativeVelocityIndicator or any combination of them;

e) shall include the lcs-QoS IE; and

g) may include the lcsClientExternalID IE indicating identity of the LCS client or the AF. If the lcsClientExternalID is included, the UE may include the mlc-Number IE indicating the GMLC via which the LCS client or the AF is to be be accessed, may include the lcsServiceTypeID IE and may include pseudonymIndicator to indicate that the pseudonym is needed.

When the network receives the lcs-SLMOLR invoke component included in the REGISTER or FACILITY message, the network shall perform the "procedures of SL-MO-LR involving LMF" as specified in 3GPP TS 23.273 [2].

If the network accepts the lcs-SLMOLR invoke component, the network shall return a FACILITY message to the UE containing lcs-SLMOLR return result component (see figure 5.2.2.9.2-1 and figure 5.2.2.9.2-2). In the lcs-SLMOLR return result component, the network:

a) may include the absoluteLocation IE, if the absoluteLocationIndicator is requested in the preferredRangingResult IE;

b) may include the absoluteVelocity IE, if the absoluteVelocityIndicator is requested in the preferredRangingResult IE;

c) may include the relativeResult IE, if the relativeLocationIndicator, rangeDirection, relativeVelocityIndicator or any combination of them are requested in the preferredRangingResult IE; and

d) may include the timestamp IE.

The use of Facility or ExtendedFacility IE in figure 5.2.2.9.2-1 and figure 5.2.2.9.2-2 are specified in clause 5.3.1.

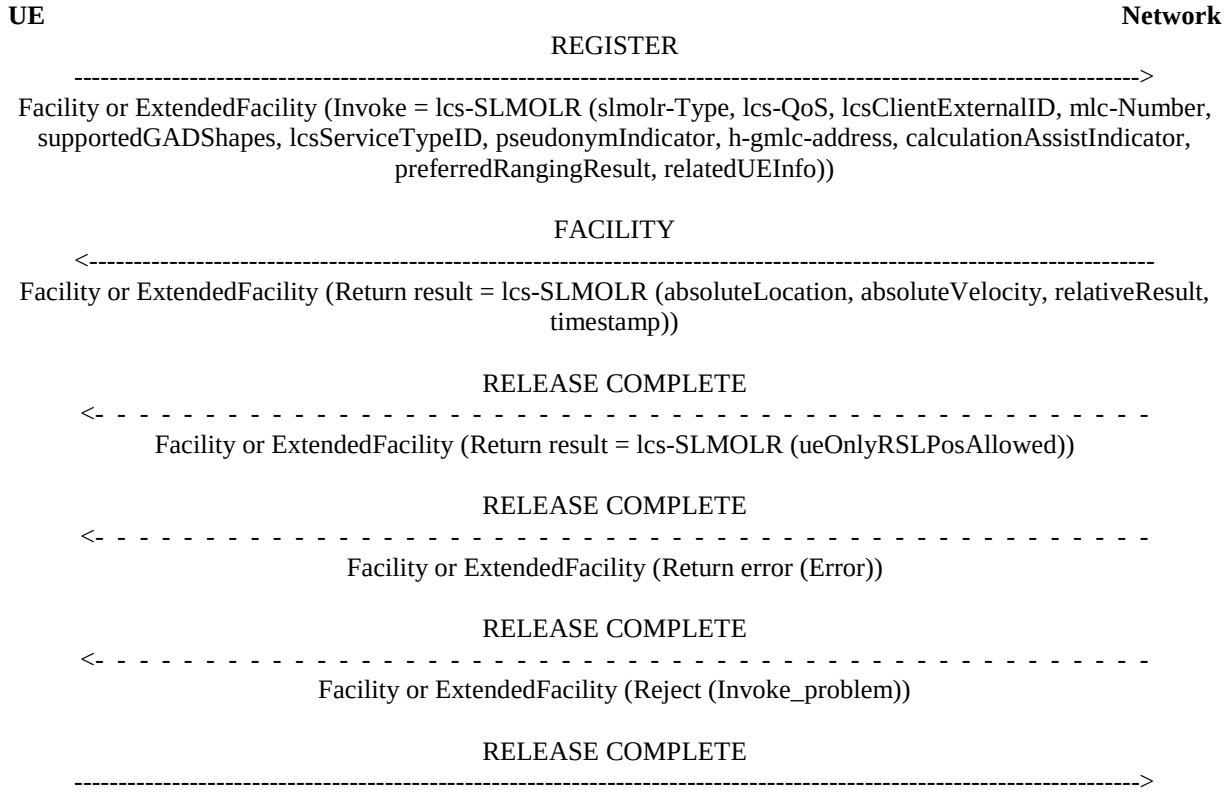


Figure 5.2.2.9.2-1: Single sidelink mobile originated location request

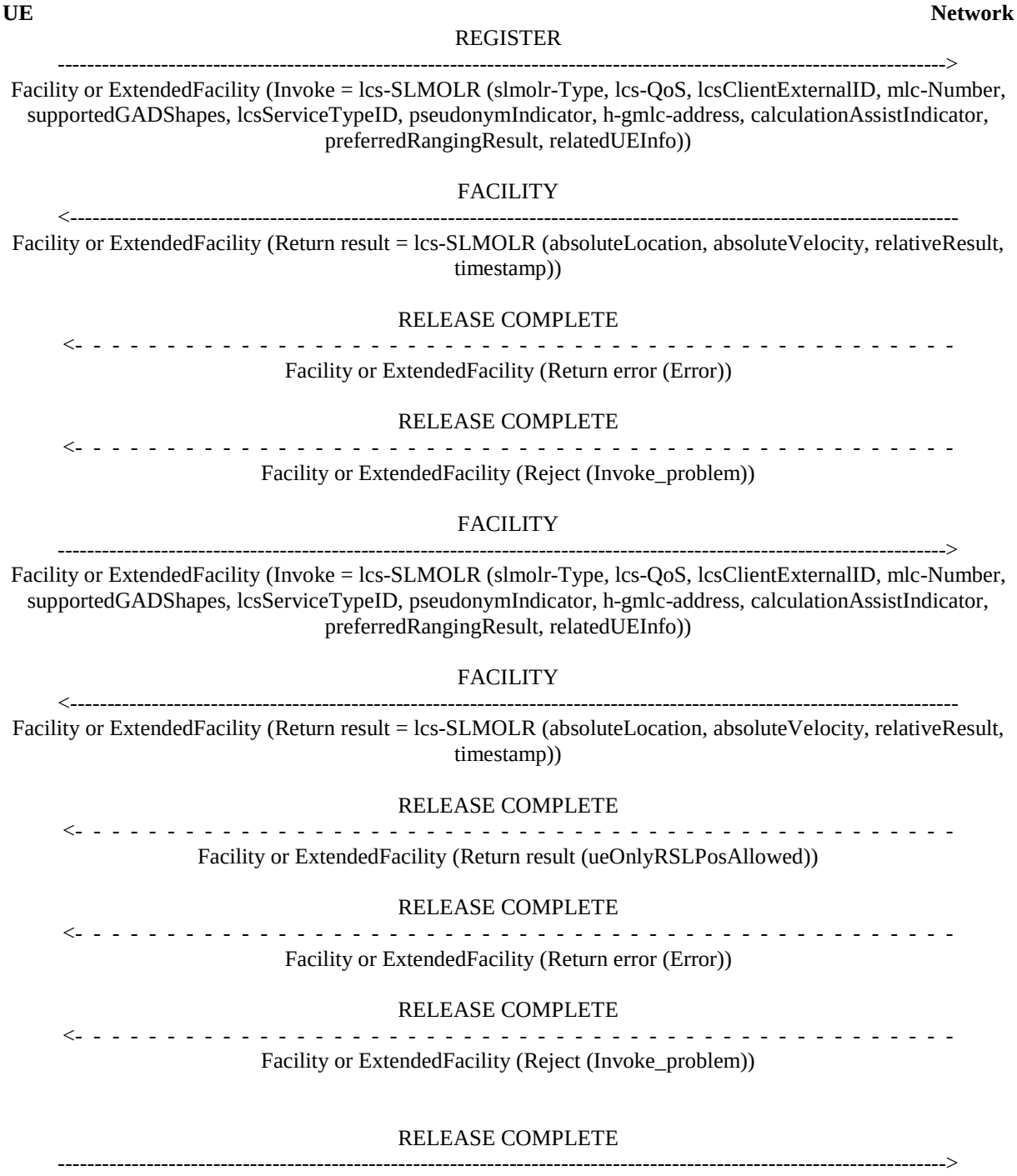


Figure 5.2.2.9.2-2: Multiple sidelink mobile originated location requests

NOTE: Only the following IEs defined in lcs-SLMOLR operation in 3GPP TS 24.080 [5] are used for ranging and sidelink positioning:

- slmolr-Type
- lcs-QoS
- lcsClientExternalID
- mlc-Number
- supportedGADShapes
- lcsServiceTypeID

- pseudonymIndicator
- h-gmlc-address
- calculationAssistIndicator
- preferredRangingResult
- relatedUEInfo
- absoluteLocation
- absoluteVelocity
- relativeResult
- ueOnlyRSLPosAllowed
- timestamp

5.2.2.10 UE initiated Sidelink Positioning Information Transport

The UE sends SLPP message and the associated Routing identifier in the UL NAS TRANSPORT message (refer to 3GPP TS 24.501 [3], 3GPP TS 23.586 [10], and 3GPP TS 23.273 [2] clause 6.20). Figure 5.2.2.10-1 illustrates an example of the NAS signalling transport for uplink SLPP messages.

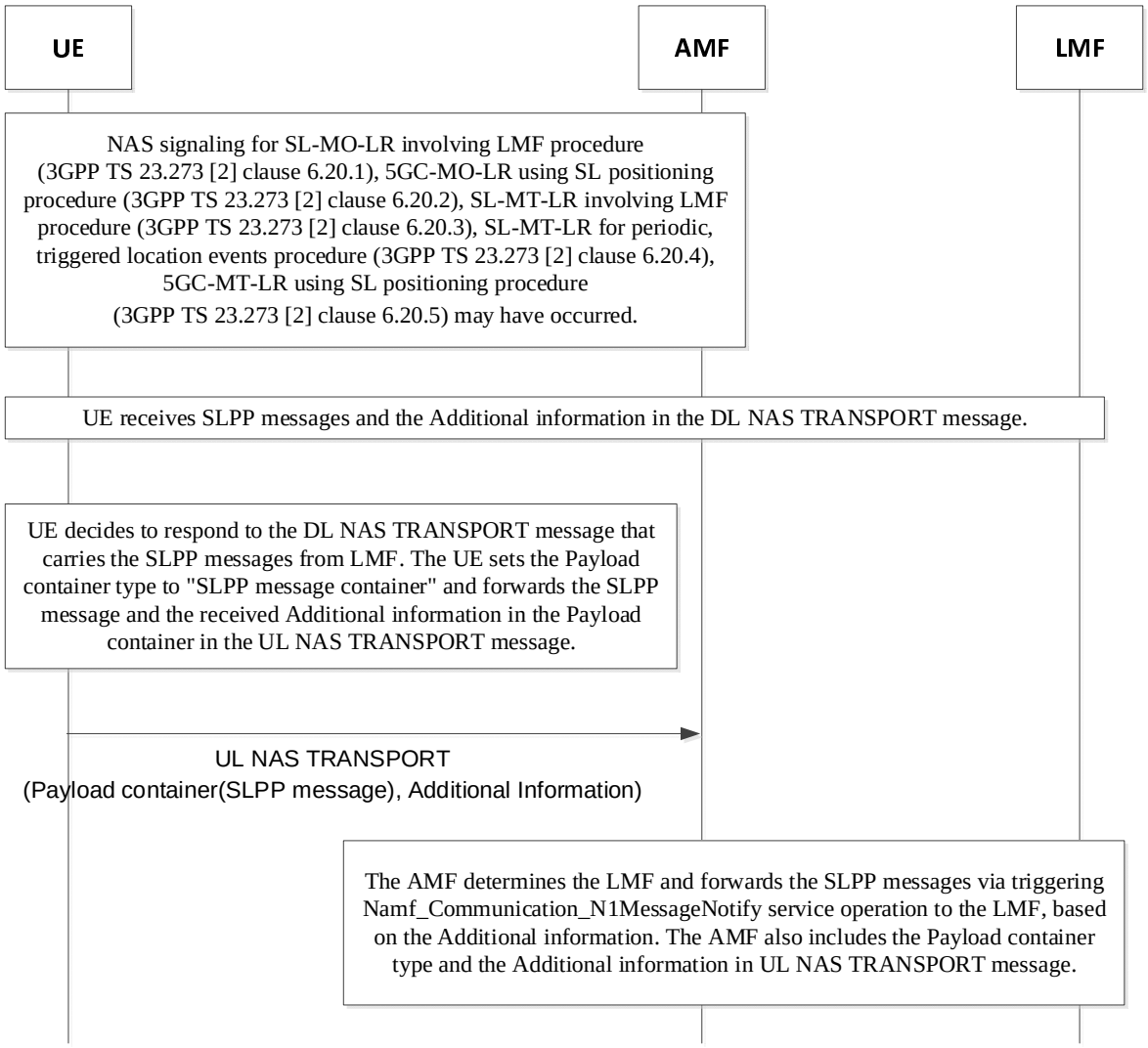


Figure 5.2.2.10-1: NAS signalling transport for uplink SLPP messages

5.2.2.11 UE initiated RSPP supplementary information transport

5.2.2.11.1 General

The UE initiated RSPP supplementary information transport procedure enables the target UE to transfer the RSPP supplementary information to the LMF for capability exchange, assistance data exchange or location information exchange as described in clause 6.20.1, clause 6.20.2, clause 6.20.3, clause 6.20.4 or clause 6.20.5 of 3GPP TS 23.273 [2].

The LCS message carrying the lcs-ULRSPPTransport invoke component is transferred from the initiating UE to the LMF via the serving AMF in an UL NAS TRANSPORT message. An acknowledgement message from the LMF is returned to the LMF via the serving AMF and is transferred to the initiating UE in a DL NAS TRANSPORT message.

5.2.2.11.2 Normal operation

In order to initiate the UE initiated RSPP supplementary information transport procedure, the initiating UE shall send a REGISTER message to the LMF containing the lcs-ULRSPPTransport invoke component as defined in 3GPP TS 24.080 [5].

In the lcs-ULRSPPTransport invoke component included in the REGISTER message, the UE shall include rangingSLPPList IE. In the rangingSLPPList, the UE:

a) may include the embedded SLPP message as defined in 3GPP TS 38.355 [4a] for the initiating UE; and

b) shall include the embedded SLPP message(s) as defined in 3GPP TS 38.355 [4a] for the related SL reference UE(s) or the related located UE(s) with the associated application layer ID(s).

NOTE: The SLPP message(s) sent by the initiating UE include the SLPP message(s) to provide capability, to request assistance data or to provide location information as defined in 3GPP TS 38.355 [4a].

The LMF shall then return a RELEASE COMPLETE message containing an lcs-ULRSPPTransport return result component (see figure 5.2.2.11.2-1) if the LMF accepts the lcs-ULRSPPTransport invoke component.

If the LMF does not accept the lcs-ULRSPPTransport invoke component, it shall return an error indication by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080 [5].

During the UE initiated RSPP supplementary information transport operation, the UE shall run a timer T. This timer is started when the operation is sent, and stopped when a response is received from the network. If this timer expires the UE shall assume that the operation has failed. The UE may terminate the dialogue by sending a RELEASE COMPLETE message, and informs the user of the failure.

Figure 5.2.2.11.2-1 illustrates the signalling for normal operation between the UE and the network. The use of Facility or ExtendedFacility IE is specified in clause 5.3.1.

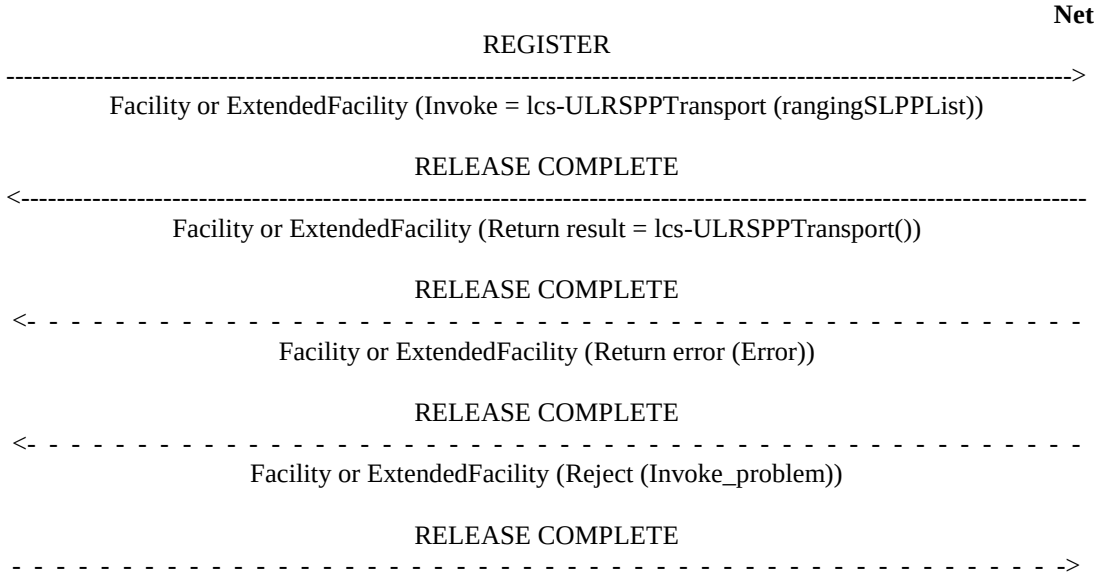


Figure 5.2.2.11.2-1: UE initiated RSPP supplementary information transport

NOTE 1: Only the following IEs defined in lcs-ULRSPPTransport operations in 3GPP TS 24.080 [5] are used for ranging and sidelink positioning:

- rangingSLPPList

NOTE 2: The rangingSLPPList IE is added to lcs-ULRSPPTransport invoke component to allow for passing multiple SLPP messages for the UE connecting with the LMF and/or the other related UE. Its ASN.1 description is given in 3GPP TS 24.080 [5].

5.3 LCS message and coding

5.3.1 Messages for Location services operations

The LCS message format and information elements coding for the lcs-MOLR, lcs-LocationNotification, lcs-EventReport, lcs-PeriodicTriggeredInvoke, lcs-CancelDeferredLocation, lcs-MSCancelDeferredLocation, lcs-LocationPrivacySetting, lcs-SLMOLR, lcs-SLMTLR, lcs-PruAssociation, lcs-PruDisassociation, lcs-DLRSPPTransport, and lcs-ULRSPPTransport operations (clause 5.2) are defined in 3GPP TS 24.080 [5] for the following messages:

- REGISTER message;
- FACILITY message; and
- RELEASE COMPLETE message.

For a component of the location services operation in clause 5.2, when the length of the component is less than or equal to 255 octets, the component shall be carried in the Facility IE of the REGISTER message, FACILITY message or RELEASE COMPLETE message, as defined in 3GPP TS 24.080 [5].

For a component of the location services operation used in clause 5.2, when the length of the component cannot be fitted into the Facility IE, i.e. exceeds 255 octets, and the UE and the network support ExtendedFacility IE for 5G LCS as specified in clause 4.2, the component shall be carried in the ExtendedFacility IE of the REGISTER message, FACILITY message or RELEASE COMPLETE message. The Facility IE shall be empty and ignored when ExtendedFacility IE is present, as defined in 3GPP TS 24.080 [5].

5.3.2 Messages for LTE Positioning Protocol (LPP)

5.3.2.1 Downlink Positioning Information Transport using LPP messages

The AMF shall set the Payload container type to "LPP message container" in the DL NAS TRANSPORT message.

The AMF includes a Routing identifier in the Additional information IE of the DL NAS TRANSPORT message which identifies the LMF and the positioning session between the AMF and LMF when a positioning session is being used.

The Routing identifier is the LCS correlation identifier, which is defined as Correlation ID in 3GPP TS 29.572 [6], so that the AMF can map the Routing identifier to the LMF and the LCS correlation identifier when the AMF receives an UL NAS TRANSPORT message including the responding LPP message.

5.3.2.2 Uplink Positioning Information Transport using LPP messages

The UE shall set the Payload container type to "LPP message container" in the UL NAS TRANSPORT message.

The UE includes a Routing identifier received in the Additional information IE of the DL NAS TRANSPORT message in the Additional information IE of the UL NAS TRANSPORT message. This association of the Routing Identifier is provided at the LPP level: the UL NAS TRANSPORT message carries an LPP message that is a response to or instigated by the LPP message in the DL NAS TRANSPORT message. The Routing identifier is the LCS correlation identifier, which is defined as Correlation ID in 3GPP TS 29.572 [6], so that the AMF can map the Routing identifier to the LCS correlation identifier when the AMF receives the UL NAS TRANSPORT message.

5.3.3 Messages for SideLink Positioning Protocol (SLPP)

5.3.3.1 Downlink Positioning Information Transport using SLPP messages

The AMF shall set the Payload container type to "SLPP message container" in the DL NAS TRANSPORT message.

The AMF includes a routing identifier in the Additional information IE of the DL NAS TRANSPORT message which identifies the LMF and the positioning session between the AMF and LMF when a positioning session is being used.

The routing identifier is the LCS correlation identifier, which is defined as Correlation ID in 3GPP TS 29.572 [6], so that the AMF can map the LCS routing identifier to the LMF and the correlation identifier when the AMF receives an UL NAS TRANSPORT message including the responding SLPP message.

5.3.3.2 Uplink Positioning Information Transport using SLPP messages

The UE shall set the Payload container type to "SLPP message container" in the UL NAS TRANSPORT message.

The UE includes a routing identifier received in the Additional information IE of the DL NAS TRANSPORT message in the Additional information IE of the UL NAS TRANSPORT message. This association of the routing identifier is provided at the SLPP level: the UL NAS TRANSPORT message carries an SLPP message that is a response to or instigated by the SLPP message in the DL NAS TRANSPORT message. The routing identifier is the LCS correlation identifier, which is defined as Correlation ID in 3GPP TS 29.572 [6], so that the AMF can map the routing identifier to the LCS correlation identifier when the AMF receives the UL NAS TRANSPORT message.

Annex A (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2019-08	CT1#119					Draft skeleton provided by the rapporteur	0.0.0
2019-10	CT1#120					Includes the following contribution agreed by CT1 at CT1#120:C1-196563	0.1.0
2019-11	CT1#121					Includes the following contributions agreed by CT1 at CT1#121:C1-198290, C1-198720	0.2.0
2019-12	CT#86	CP-193155				Presentation for information to TSG CT	1.0.0
2020-03	CT#87e	CP-200287				Implementation of pCRs CP-200102 and 103 approved by TSG CT. Presentation for approval to TSG CT.	2.0.0
2020-03	CT#87e					Version 16.0.0 created after approval	16.0.0
2020-03	CT#87e					Editorial corrections	16.0.1
2020-06	CT#88e	CP-201098	0001		B	Adding Location Privacy Setting operation	16.1.0
2020-09	CT#89e	CP-202148	0002	3	B	UE initiated Event Reporting Procedure for Low Power Event Reporting	16.2.0
2020-09	CT#89e	CP-202148	0003		F	Additional function of MO-LR procedure	16.2.0
2021-12	CT#84e	CP-213031	0004	1	F	TS reference update and multiplePositioningProtocolPDUs limit clarification	17.0.0
2022-03	CT#95e	CP-220243	0006	1	B	AMF LCS functionality for satellite access	17.1.0
2022-03	CT#95e	CP-220243	0008	1	F	Clarification on multiplePositioningProtocolPDUs IE	17.1.0
2022-03	CT#95e	CP-220243	0009	4	F	Clarification on the LMF ID	17.1.0
2022-03	CT#95e	CP-220243	0010	1	F	Clarification on Routing information	17.1.0
2022-06	CT#96	CP-221207	0011	1	F	The Location Service partially applicable for SNPN	17.2.0
2022-06	CT#96	CP-221207	0012	1	B	Additional of Scheduled Location Time	17.2.0
2022-06	CT#96	CP-221207	0013	1	F	Handling of Scheduled Location Time by UE	17.2.0
2022-12	CT#98e	CP-223115	0014	1	F	Correction on country verification for satellite access	17.3.0
2023-03	CT#99	CP-230210	0015	3	B	Signalling support to enable a periodic or triggered 5GC-MT-LR event reporting over user plane connection	18.0.0
2023-03	CT#99	CP-230285	0016		F	Several corrections on the reference to 23.273	18.0.0
2023-03	CT#99	CP-230210	0017	2	B	NAS protocol supports LCS PRU messages	18.0.0
2023-03	CT#99	CP-230210	0018	1	B	Enhance the Triggered Location for UE power saving purpose	18.0.0
2023-06	CT#100	CP-231207	0019	2	B	New procedures for PRU UE	18.1.0
2023-06	CT#100	CP-231207	0030	-	F	Correction to the IE's name and the correlation NOTE	18.1.0
2023-06	CT#100	CP-231207	0025	2.0	B	Enhancement for location service continuity from 5GS to EPS	18.1.0
2023-06	CT#100	CP-231207	0027	2	B	Restriction on NR satellite access for PRU UE	18.1.0
2023-06	CT#100	CP-231207	0031	1	B	Enhancement of the event report allowed area	18.1.0
2023-06	CT#100	CP-231207	0028	3	B	Definition of PRU UE	18.1.0
2023-06	CT#100	CP-231207	0026	4	F	Edit misaligned note	18.1.0
2023-06	CT#100	CP-231207	0032	2	B	Addition of EPC-(H)GMLC address in LCS-PeriodicTriggered	18.1.0
2023-09	CT#101	CP-232201	0034	-	F	Update the defination of UE operating as PRU	18.2.0
2023-09	CT#101	CP-232201	0041	1	F	Update the terminology of PRU UE into UE operating as PRU	18.2.0
2023-09	CT#101	CP-232201	0039	1	F	Add reporting indication to Supplementary Services Periodic or Triggered Location	18.2.0
2023-09	CT#101	CP-232201	0040	2	B	Add reporting indication to UE initiated Event Reporting Procedure	18.2.0
2023-12	CT#102	CP-233143	0046	-	D	Editorial correction for MO-LR	18.3.0
2023-12	CT#102	CP-233128	0049	-	B	PRU ON/OFF state	18.3.0
2023-12	CT#102	CP-233128	0051	1	F	Addition of parameters in PRU procedure	18.3.0
2023-12	CT#102	CP-233128	0052	-	F	Solve EN in PRU disassociation procedure	18.3.0
2023-12	CT#102	CP-233182	0044	1	B	Additional of the SL-MO-LR procedure	18.3.0
2023-12	CT#102	CP-233128	0056	-	F	Correction to reference	18.3.0
2023-12	CT#102	CP-233128	0054	1	F	Payload container information IE related to PRU	18.3.0
2023-12	CT#102	CP-233182	0047	3	B	NAS transport for RSPP	18.3.0
2023-12	CT#102	CP-233182	0055	1	B	SLPP introduction	18.3.0
2024-03	CT#103	CP-240087	0058	-	D	Typo corrections	18.4.0
2024-03	CT#103	CP-240120	0064	-	F	Correction to NAS signalling transport for downlink SLPP messages	18.4.0
2024-03	CT#103	CP-240120	0059	1	B	SL-MO-LR response in case of network congestion	18.4.0
2024-03	CT#103	CP-240087	0065	1	D	Miscellaneous corrections	18.4.0
2024-03	CT#103	CP-240087	0066	1	F	Update the positioning information transport procedure for PRU	18.4.0
2024-03	CT#103	CP-240087	0068	2	F	Clarifications for reporting indication	18.4.0
2024-03	CT#103	CP-240120	0069	1	B	Procedure for Ranging/SL Positioning service exposure through 5GC network via control plane	18.4.0
2024-03	CT#103	CP-240120	0071	2	B	UL Supplementary service message between LMF and UE for Ranging_SL	18.4.0
2024-03	CT#103	CP-240120	0070	2	B	DL Supplementary service message between LMF and UE for Ranging_SL	18.4.0
2024-03	CT#103	CP-240120	0043	6	B	Mobile Terminated Location Request for Ranging_SL	18.4.0
2024-06	CT#104	CP-241154	0075	-	F	Clarification on the parameters for the UP and cumulative event reporting	18.5.0
2024-06	CT#104	CP-241154	0076	-	F	Clarification on PRU capability	18.5.0
2024-06	CT#104	CP-241192	0073	1	F	Removal of ranging SL positioning service exposure through 5GC network via control plane	18.5.0

2024-06	CT#104	CP-241192	0077	1	F	Update of rangingResult IE	18.5.0
2024-06	CT#104	CP-241153	0081	-	F	Revise the PRU disassociation procedure	18.5.0
2024-06	CT#104	CP-241192	0079	1	F	Update overview clause to capture ranging operations	18.5.0
2024-06	CT#104	CP-241192	0080	1	F	Notification for privacy check on UE for RangingSl	18.5.0
2024-06	CT#104					Editorial	18.5.1
2024-09	CT#105	CP-242193	0083	-	F	Correction to LCS message used in SL-MT-LR procedure	18.6.0
2024-09	CT#105	CP-242206	0087	1	F	Clarification on the positioning IEs	18.6.0
2024-09	CT#105	CP-242193	0084	1	F	Corrections to RSPP supplementary information transport procedure	18.6.0
2024-09	CT#105	CP-242193	0086	1	F	Correct the foramt of the NOTE for the Ranging IEs	18.6.0
2024-09	CT#105	CP-242193	0085	1	F	Correction to the NOTE in normal operation for SL-MO-LR	19.0.0
2024-12	CT#106	CP-243195	0092	-	A	Information elements in SL-MT-LR response	19.1.0
2024-12	CT#106	CP-243200	0095	1	F	Correction on AMF sending LocationPrivacySetting return result to the UE	19.1.0
2024-12	CT#106	CP-243186	0088	1	F	24.571 clean up	19.1.0
2024-12	CT#106	CP-243195	0090	2	F	Update on SL-MT-LR for periodic, triggered Location Events	19.1.0
2024-12	CT#106	CP-243176	0096	2	F	Alignments of parameters for LCS-SS messages	19.1.0
2024-12	CT#106	CP-243195	0093	1	F	Corrections to the UL NAS TRANSPORT and DL NAS TRANSPORT messages	19.1.0
2024-12	CT#106	CP-243195	0094	1	F	Correction to the inconsistent correlation identifier	19.1.0
2024-12	CT#106	CP-243176	0097	1	D	Editorial corrections on 24.571	19.1.0
2024-12	CT#106	CP-243206	0099	1	A	Correction to missing message RELEASE COMPLETE for network initiated location service operations.	19.1.0
2024-12	CT#106	CP-243206	0101	4	A	Correction to missing RELEASE COMPLETE for UE initiated location service operations.	19.1.0
2025-09	CT#109	CP-252155	0102	1	D	LCS session identifiers	19.2.0
2025-12	CT#110	CP-253114	0105	1	F	Routing ID per LCS session	19.3.0
2026-03	CT#111	CP-260125	0108		F	Correction of non-existent lcs-SL-MTLR operation	19.4.0
2026-03	CT#111	CP-260125	0109	2	F	Adding support for ExtendedFacility IE for 5G LCS	19.4.0
2026-09	CT#113	CP-262062	0111	1	F	Support Location Service Continuity for deferred location – EPS to 5GS	19.5.0
2026-09	CT#113	CP-262062	0112	1	F	Cancel Deferred Location operation applies for EPC-MT-LR procedure from EPS	19.5.0
2026-09	CT#113	CP-262062	0113	1	F	Location Service Continuity for deferred routing id	19.5.0

History

Version	Date	Status
V19.2.0	January 2026	Publication
V19.3.0	February 2026	Publication
V19.4.0	March 2026	Publication
V19.5.0	September 2026	Publication